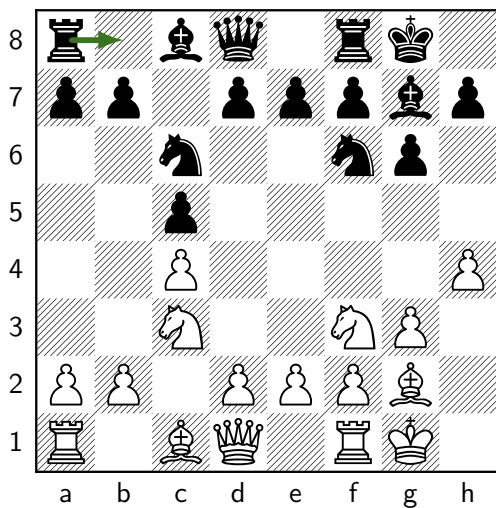


Melanie-XRAY — English, Symmetrical

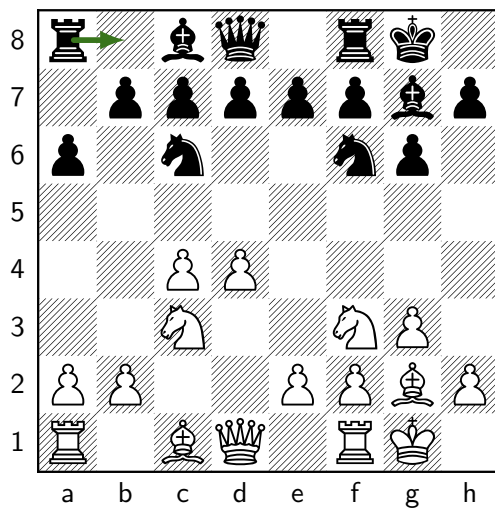
Melanie (Deep_Vision)

Abstract

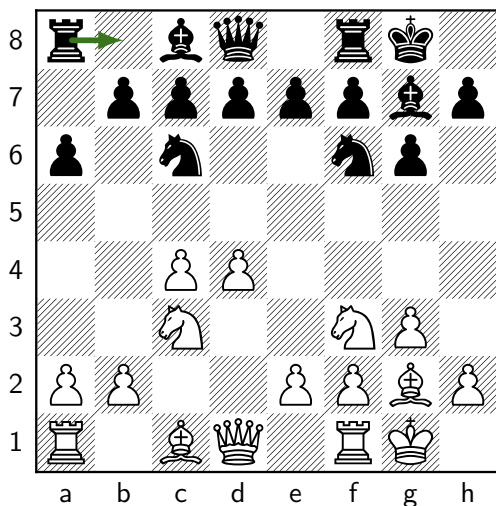
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie's move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under μ – cross-type maps (a rook playing a queen's role) are called out, as no material/eval engine explains a move that way.



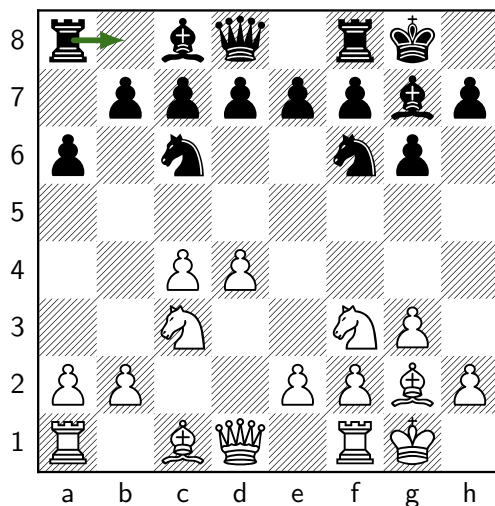
(a)



(b)

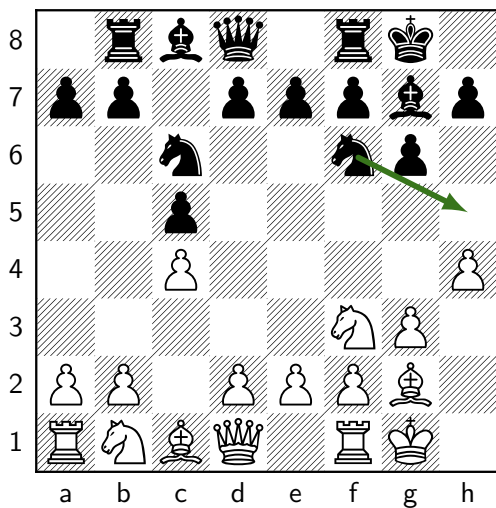


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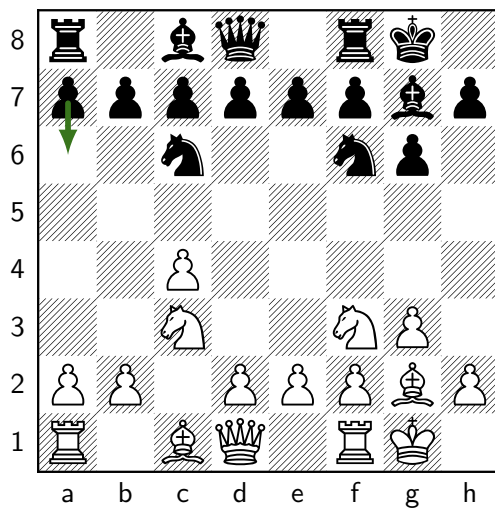


(d)

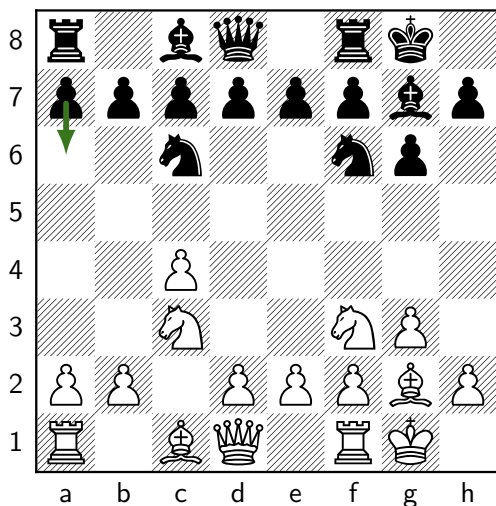
Figure 1: **Move 7, Black: Rb8.** (a) Our position; Melanie plays Rb8 (a8b8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1805` (ply 13) [structural distance $d=22$, QAP obj=14]: the master played Rb8; under μ our *Ra1* plays the role of that game’s *Ra1*; cross-type roles: our *pe2* as their *Qd8*, our *Bc8* as their *Pc7*; transferred move a8b8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1806` (ply 13) [structural distance $d=22$, QAP obj=14]: the master played Rb8; under μ our *Ra1* plays the role of that game’s *Ra1*; cross-type roles: our *pe2* as their *Qd8*, our *Bc8* as their *Pc7*; transferred move a8b8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3290` (ply 13) [structural distance $d=22$, QAP obj=14]: the master played Rb8; under μ our *Ra1* plays the role of that game’s *Ra1*; cross-type roles: our *pe2* as their *Qd8*, our *Bc8* as their *Pc7*; transferred move a8b8 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Rb8 (summed QAP objective 102, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



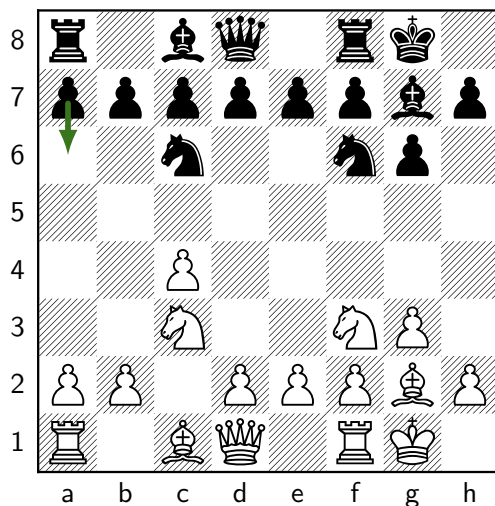
(a)



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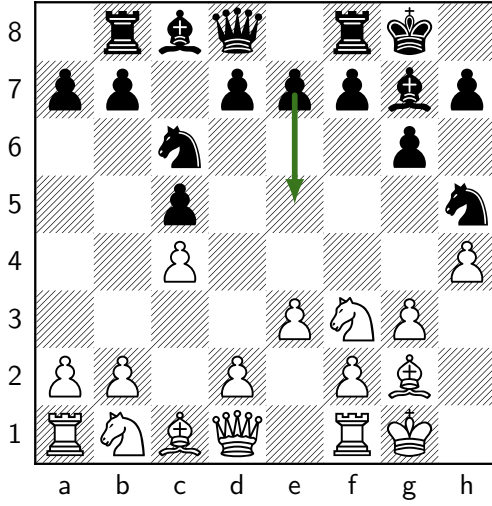


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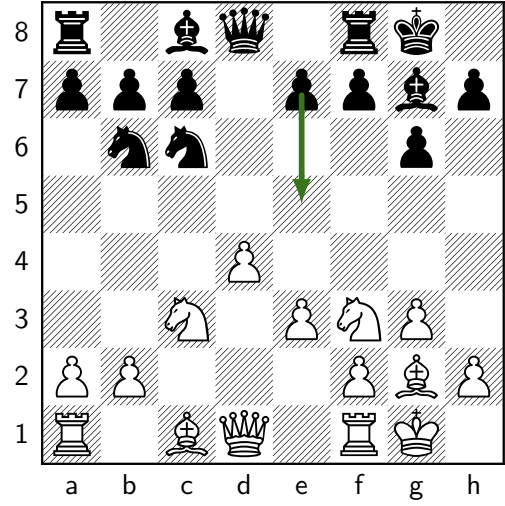


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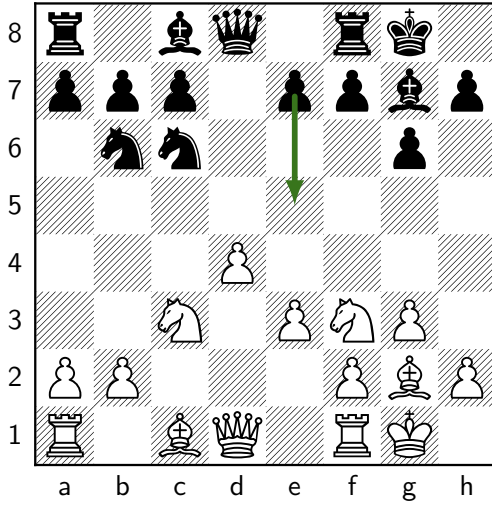
Figure 2: **Move 8, Black: Nh5**. (a) Our position; Melanie plays Nh5 (f6h5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#1805` (ply 11) [structural distance $d=40$, QAP obj=10]: the master played a6; under μ our $Nf3$ plays the role of that game’s $Pa2$ (a **cross-type** correspondence); cross-type roles: our $pf2$ as their $Bc8$, our $nf3$ as their $Pb7$; transferred move f6h5 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#1806` (ply 11) [structural distance $d=40$, QAP obj=10]: the master played a6; under μ our $Nf3$ plays the role of that game’s $Pa2$ (a **cross-type** correspondence); cross-type roles: our $pf2$ as their $Bc8$, our $nf3$ as their $Pb7$; transferred move f6h5 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#3290` (ply 11) [structural distance $d=40$, QAP obj=10]: the master played a6; under μ our $Nf3$ plays the role of that game’s $Pa2$ (a **cross-type** correspondence); cross-type roles: our $pf2$ as their $Bc8$, our $nf3$ as their $Pb7$; transferred move f6h5 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Nh5 (summed QAP objective 41, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



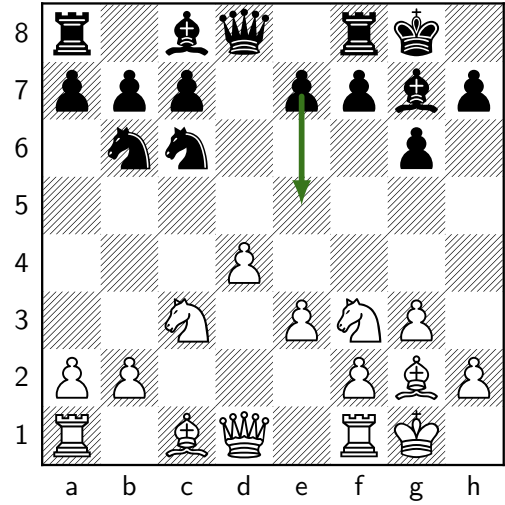
(a)



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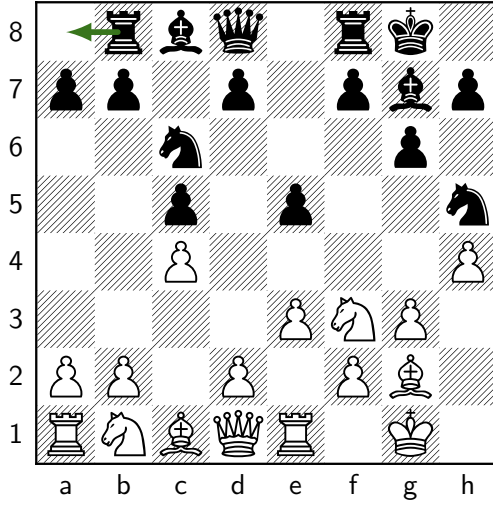


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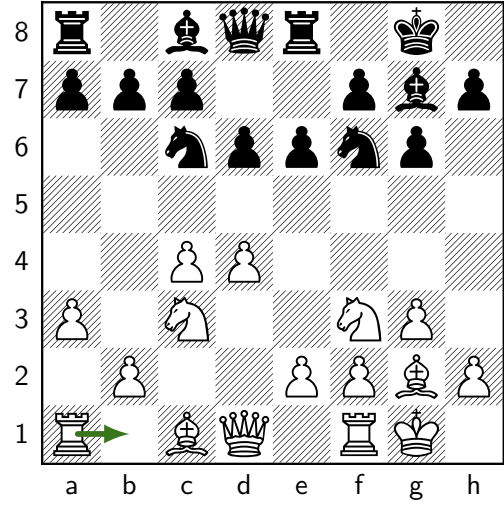


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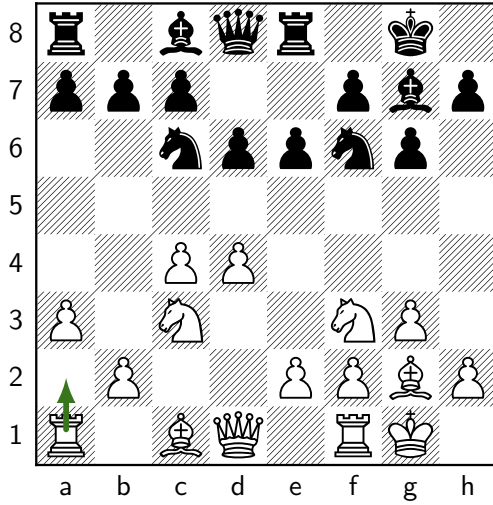
Figure 3: **Move 9, Black: e5.** (a) Our position; Melanie plays e5 (e7e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#2647` (ply 17) [structural distance $d=37$, QAP obj=59]: the master played e5; under μ our $Pe2$ plays the role of that game’s $Pe2$; cross-type roles: our $Nh5$ as their $Bc8$, our $Bg7$ as their $Qd8$; transferred move e7e5 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#3536` (ply 17) [structural distance $d=37$, QAP obj=59]: the master played e5; under μ our $Pe2$ plays the role of that game’s $Pe2$; cross-type roles: our $Nh5$ as their $Bc8$, our $Bg7$ as their $Qd8$; transferred move e7e5 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#1240` (ply 17) [structural distance $d=37$, QAP obj=59]: the master played e5; under μ our $Pe2$ plays the role of that game’s $Pe2$; cross-type roles: our $Nh5$ as their $Bc8$, our $Bg7$ as their $Qd8$; transferred move e7e5 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to e5 (summed QAP objective 394, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



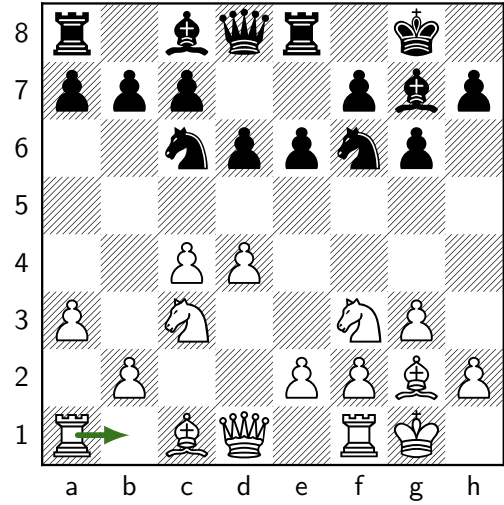
(a)



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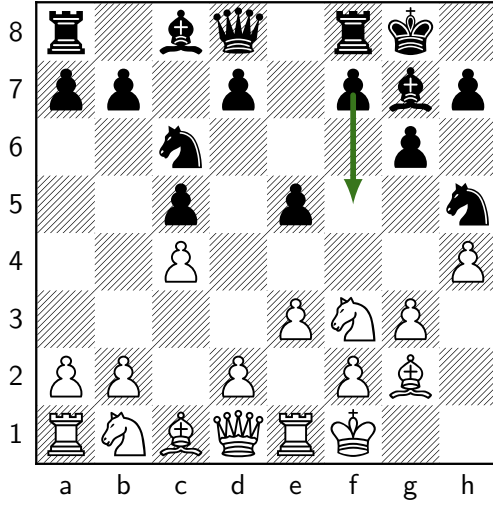


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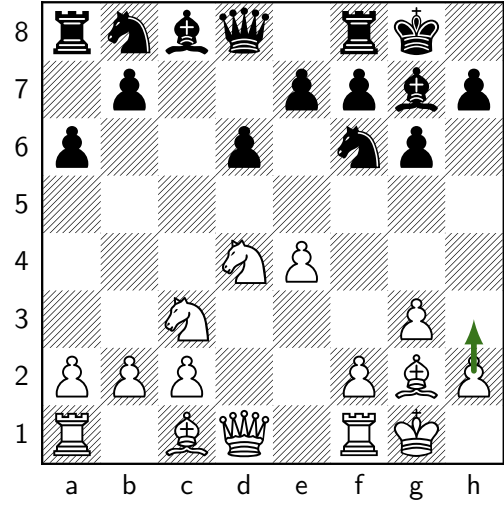


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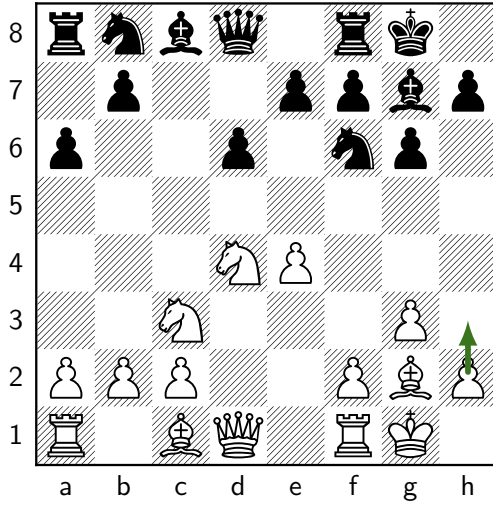
Figure 4: **Move 10, Black: Ra8.** (a) Our position; Melanie plays Ra8 (b8a8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#808` (ply 16) [structural distance $d=42$, QAP obj=29]: the master played Rb1; under μ our Rb1 plays the role of that game’s Ra1; cross-type roles: our Nh5 as their Bc1, our Bc8 as their Pb2; transferred move b8a8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#809` (ply 16) [structural distance $d=42$, QAP obj=29]: the master played Ra2; under μ our Rb1 plays the role of that game’s Ra1; cross-type roles: our Nh5 as their Bc1, our Bc8 as their Pb2; transferred move b8a8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2678` (ply 16) [structural distance $d=42$, QAP obj=29]: the master played Rb1; under μ our Rb1 plays the role of that game’s Ra1; cross-type roles: our Nh5 as their Bc1, our Bc8 as their Pb2; transferred move b8a8 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Ra8 (summed QAP objective 145, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



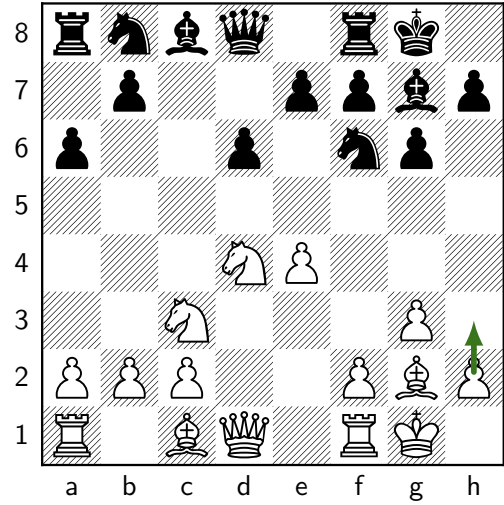
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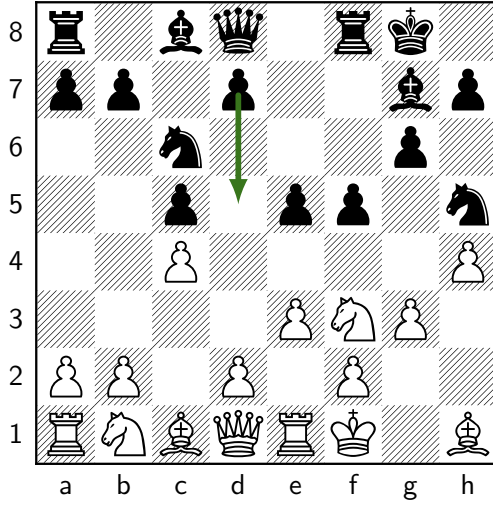


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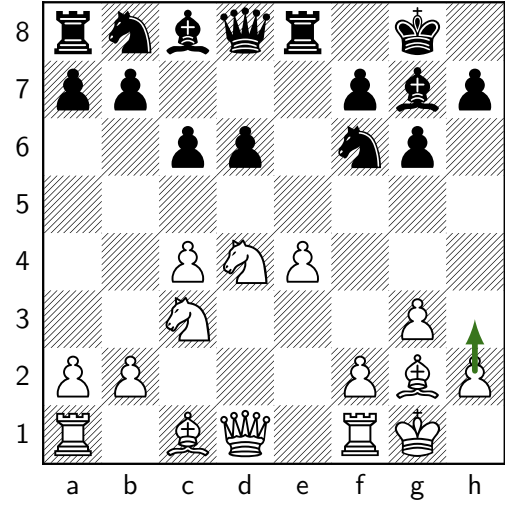


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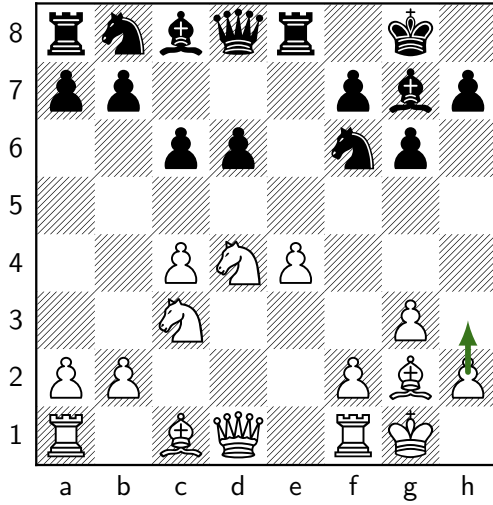
Figure 5: **Move 11, Black: f5.** (a) Our position; Melanie plays f5 (f7f5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#845` (ply 20) [structural distance $d=50$, QAP obj=38]: the master played h3; under μ our Pf2 plays the role of that game’s Ph2; cross-type roles: our Bg7 as their Pf2, our Ph7 as their Bg2; transferred move f7f5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1694` (ply 20) [structural distance $d=50$, QAP obj=38]: the master played h3; under μ our Pf2 plays the role of that game’s Ph2; cross-type roles: our Bg7 as their Pf2, our Ph7 as their Bg2; transferred move f7f5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#11` (ply 16) [structural distance $d=50$, QAP obj=38]: the master played h3; under μ our Pf2 plays the role of that game’s Ph2; cross-type roles: our Bg7 as their Pf2, our Ph7 as their Bg2; transferred move f7f5 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to f5 (summed QAP objective 190, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



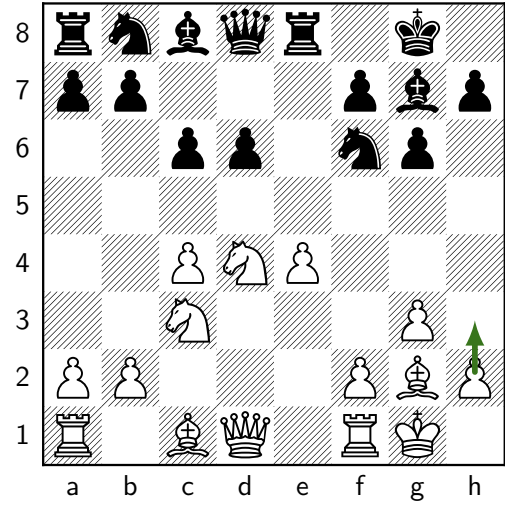
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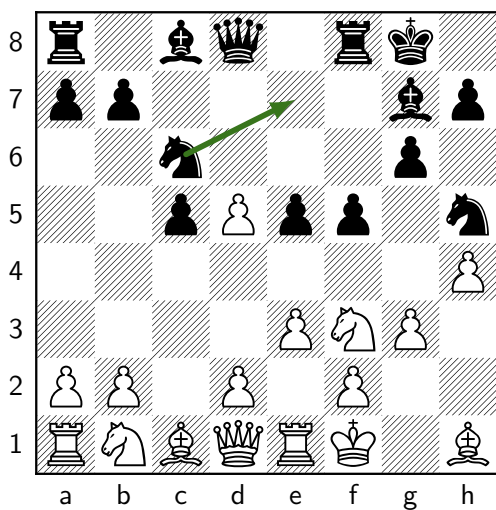


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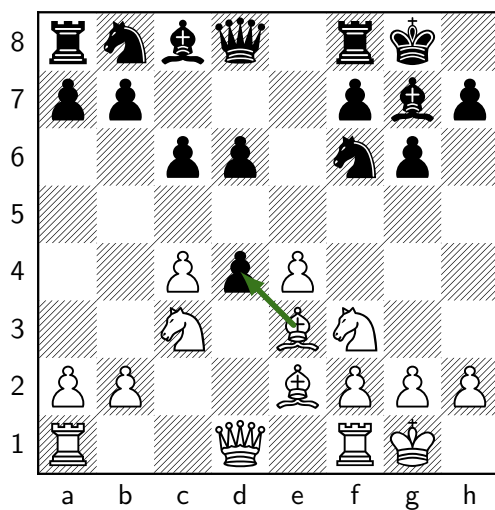


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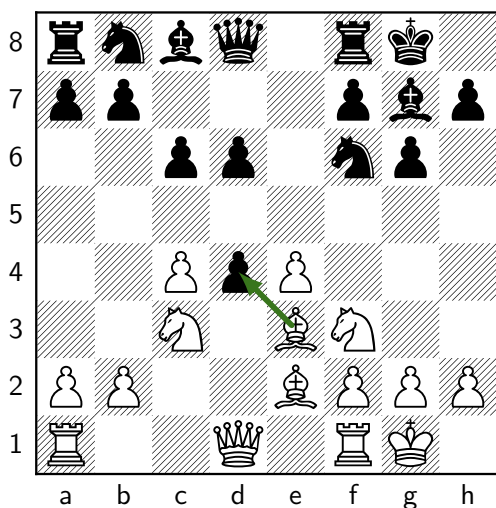
Figure 6: **Move 12, Black: d5.** (a) Our position; Melanie plays d5 (d7d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#352` (ply 18) [structural distance $d=62$, QAP obj=63]: the master played h3; under μ our $Pd2$ plays the role of that game’s $Ph2$; cross-type roles: our $bh1$ as their $pf7$, our $re1$ as their $bg7$; transferred move d7d5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#353` (ply 18) [structural distance $d=62$, QAP obj=63]: the master played h3; under μ our $Pd2$ plays the role of that game’s $Ph2$; cross-type roles: our $bh1$ as their $pf7$, our $re1$ as their $bg7$; transferred move d7d5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1243` (ply 18) [structural distance $d=62$, QAP obj=63]: the master played h3; under μ our $Pd2$ plays the role of that game’s $Ph2$; cross-type roles: our $bh1$ as their $pf7$, our $re1$ as their $bg7$; transferred move d7d5 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to d5 (summed QAP objective 252, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



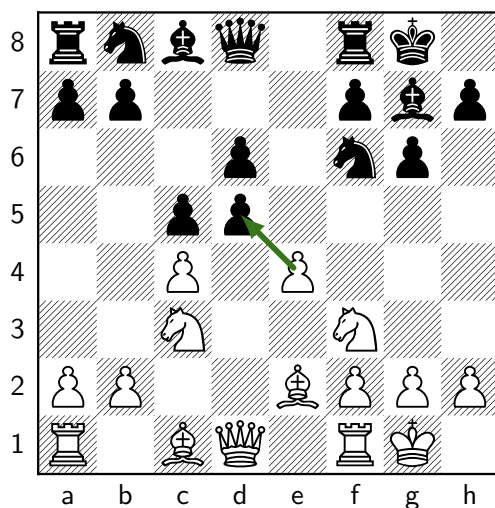
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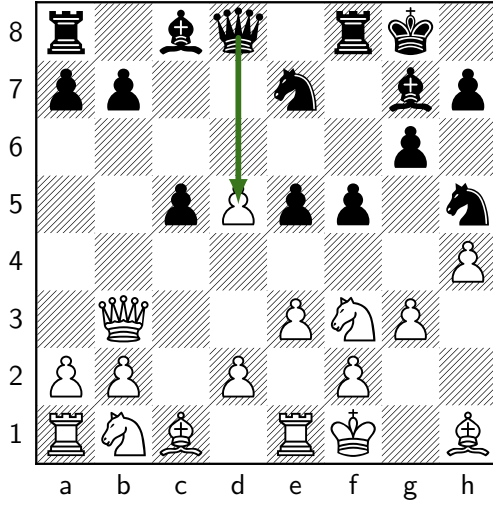


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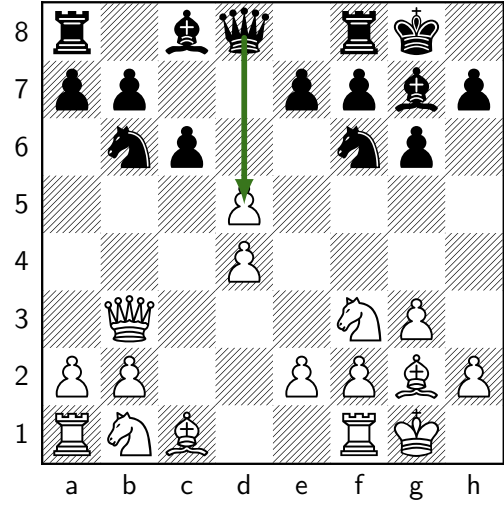


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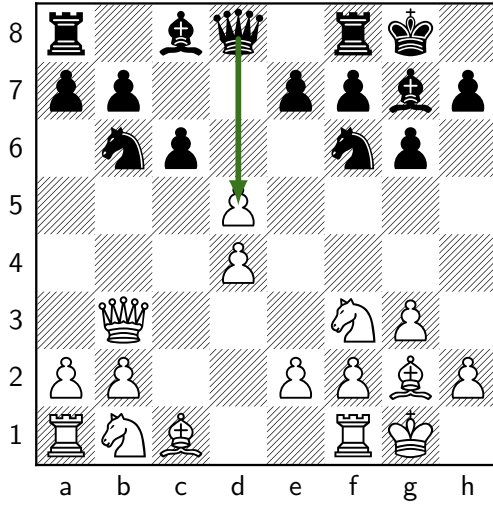
Figure 7: **Move 13, Black: Ne7.** (a) Our position; Melanie plays Ne7 (c6e7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#3423` (ply 16) [structural distance $d=72$, QAP obj=86]: the master played Bxd4; under μ our Nc3 plays the role of that game’s Be3 (a **cross-type** correspondence); cross-type roles: our Bc8 as their Ra1, our bh1 as their Pf2; transferred move c6e7 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#3424` (ply 16) [structural distance $d=72$, QAP obj=86]: the master played Bxd4; under μ our Nc3 plays the role of that game’s Be3 (a **cross-type** correspondence); cross-type roles: our Bc8 as their Ra1, our bh1 as their Pf2; transferred move c6e7 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#1610` (ply 16) [structural distance $d=76$, QAP obj=45]: the master played exd5; under μ our Nc3 plays the role of that game’s Pe4 (a **cross-type** correspondence); cross-type roles: our nf3 as their Bc1, our ph4 as their Be2; transferred move c6e7 (**endorses this move**). Synthesis: 30 retrieved analogs transferred to Ne7 (summed QAP objective 1001, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



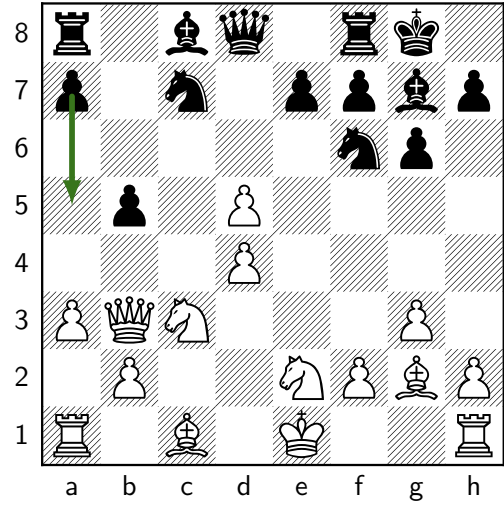
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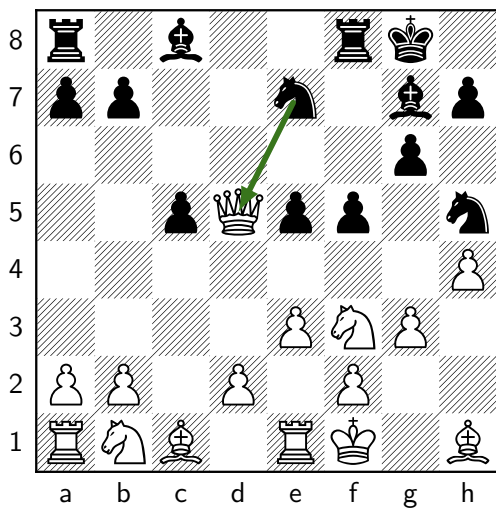


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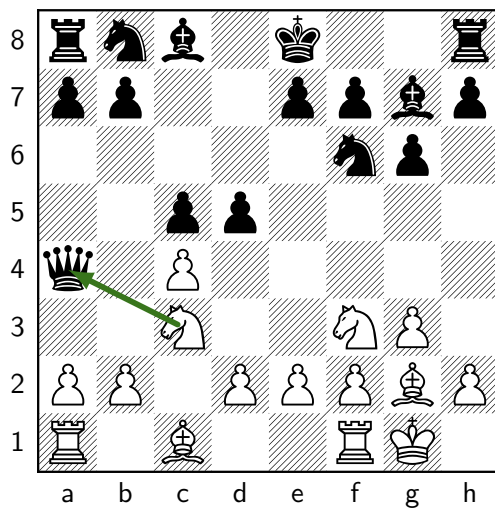


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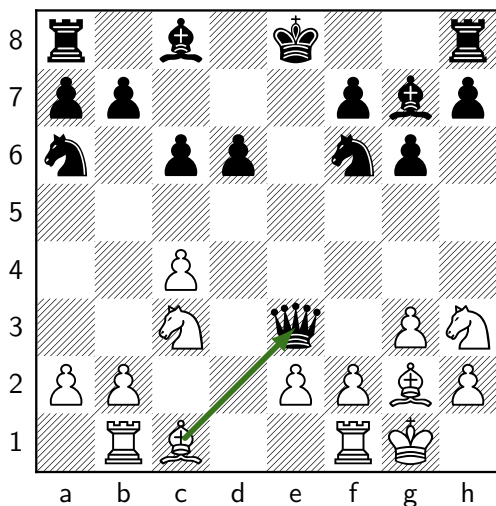
Figure 8: **Move 14, Black: Qxd5.** (a) Our position; Melanie plays Qxd5 (d8d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#737` (ply 17) [structural distance $d=87$, QAP obj=61]: the master played Qxd5; under μ our Qd1 plays the role of that game’s Qd1; cross-type roles: our Pf5 as their Bc8, our Bc8 as their Pe7; transferred move d8d5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#738` (ply 17) [structural distance $d=87$, QAP obj=61]: the master played Qxd5; under μ our Qd1 plays the role of that game’s Qd1; cross-type roles: our Pf5 as their Bc8, our Bc8 as their Pe7; transferred move d8d5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#288` (ply 21) [structural distance $d=89$, QAP obj=112]: the master played a5; under μ our Pa2 plays the role of that game’s Pa2; cross-type roles: our qb3 as their pd4, our nf3 as their qb3; transferred move a7a5. Synthesis: 2 retrieved analogs transferred to Qxd5 (summed QAP objective 122, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



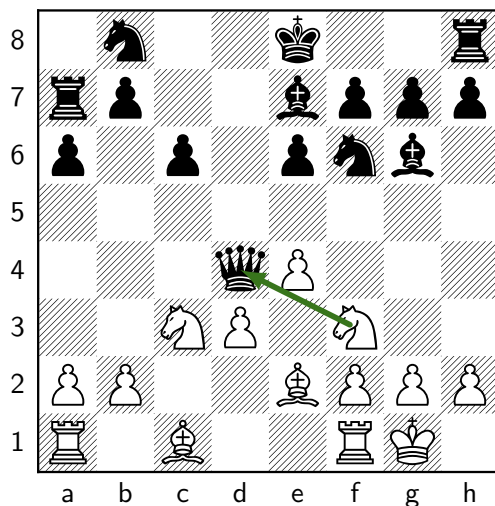
(a)



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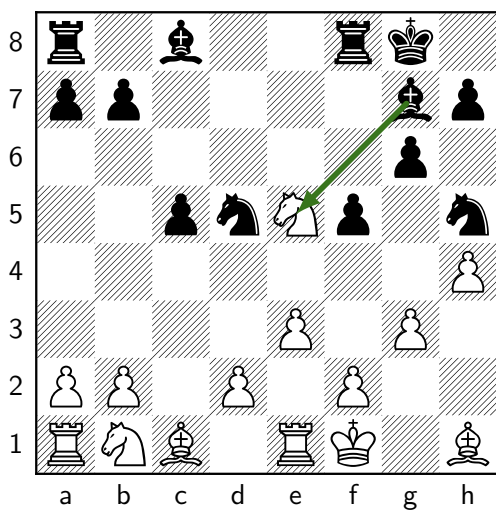


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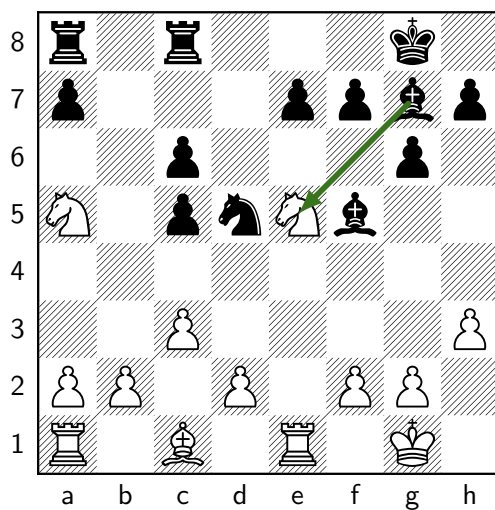


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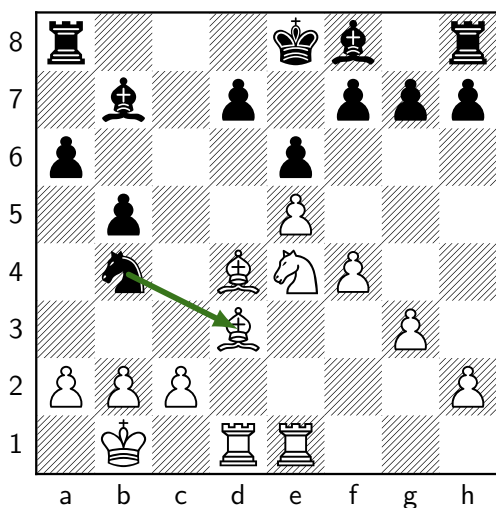
Figure 9: **Move 15, Black: Nxd5.** (a) Our position; Melanie plays Nxd5 (e7d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#218` (ply 14) [structural distance $d=114$, QAP obj=50]: the master played Nxa4; under μ our Ne2 plays the role of that game's Nc3; cross-type roles: our ra1 as their Bc1, our Rf8 as their Pb2; transferred move e7d5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3631` (ply 20) [structural distance $d=134$, QAP obj=49]: the master played Bxe3; under μ our Ne2 plays the role of that game's Bc1 (a **cross-type** correspondence); cross-type roles: our bc1 as their Rb1, our pf2 as their Bg2; transferred move e7d5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#72` (ply 22) [structural distance $d=67$, QAP obj=47]: the master played Nxd4; under μ our Ne2 plays the role of that game's Nf3; cross-type roles: our Pg6 as their Ra1, our Nh5 as their Bc1; transferred move e7d5 (**endorses this move**). Synthesis: 12 retrieved analogs transferred to Nxd5 (summed QAP objective 402, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



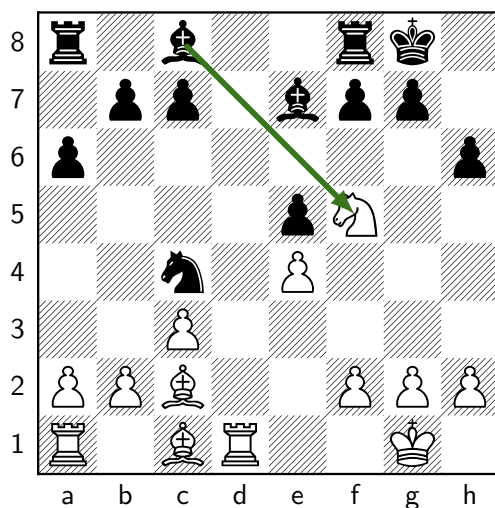
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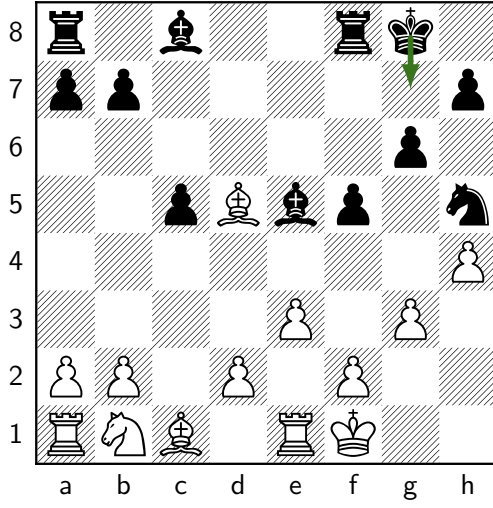


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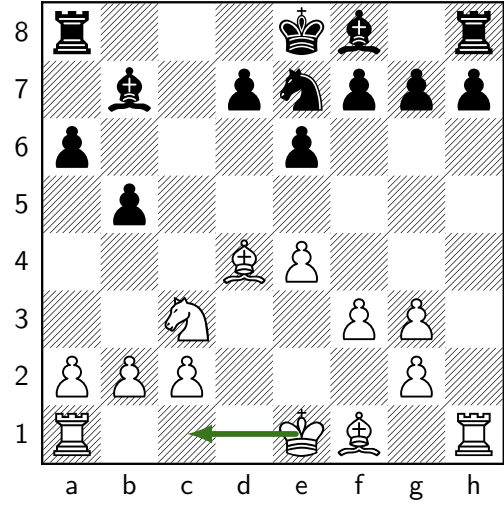


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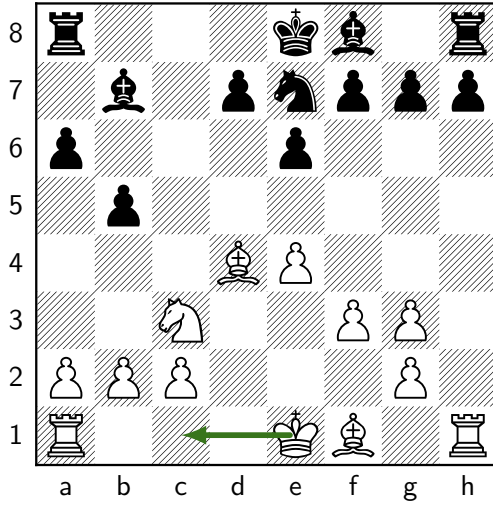
Figure 10: **Move 16, Black: Bxe5.** (a) Our position; Melanie plays Bxe5 (g7e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2221` (ply 35) [structural distance $d=80$, QAP obj=88]: the master played Bxe5; under μ our Bg2 plays the role of that game’s Bg2; cross-type roles: our Pf5 as their Ra8, our Bc8 as their Pa7; transferred move g7e5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3236` (ply 31) [structural distance $d=81$, QAP obj=57]: the master played Nx d3; under μ our Bg2 plays the role of that game’s Nb5 (a **cross-type** correspondence); cross-type roles: our Nh5 as their Bf8, our Pf5 as their Rh8; transferred move g7e5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2653` (ply 35) [structural distance $d=82$, QAP obj=52]: the master played Bxf5; under μ our Bg2 plays the role of that game’s Bc1; cross-type roles: our Bc8 as their Pb7, our Pg6 as their Be7; transferred move g7e5 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Bxe5 (summed QAP objective 197, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



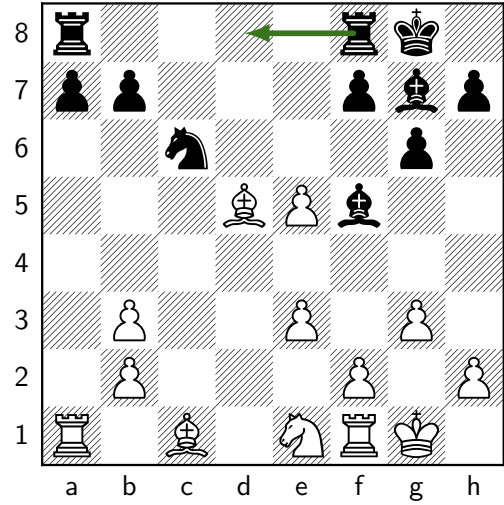
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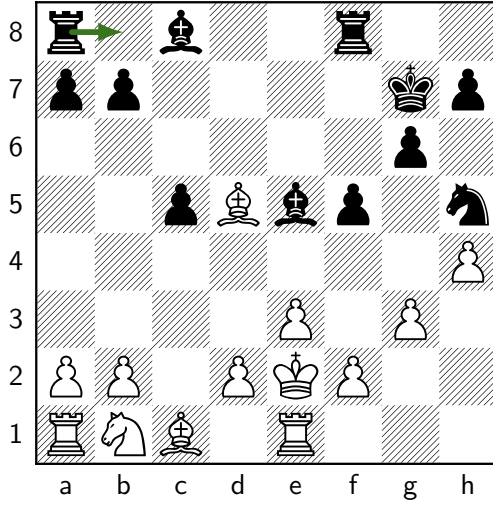


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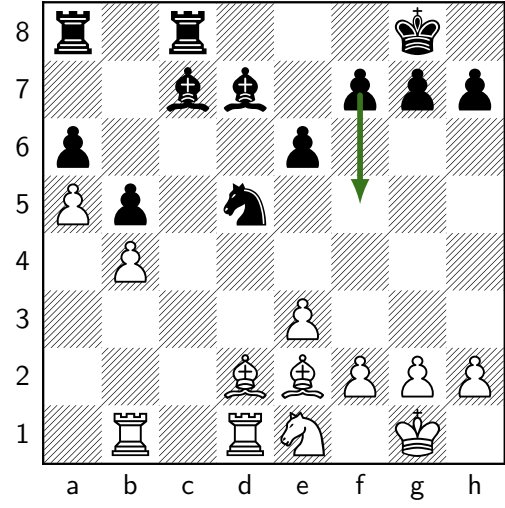


(d)

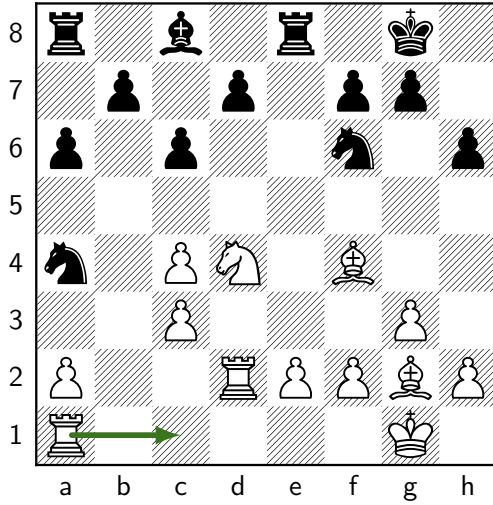
Figure 11: **Move 17, Black: Kg7.** (a) Our position; Melanie plays Kg7 (g8g7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1370` (ply 22) [structural distance $d=65$, QAP obj=26]: the master played 0-0-0; under μ our Kg1 plays the role of that game’s Ke1; cross-type roles: our Pg6 as their Bf1, our bd5 as their Rh1; transferred move g8g7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1485` (ply 22) [structural distance $d=65$, QAP obj=26]: the master played 0-0-0; under μ our Kg1 plays the role of that game’s Ke1; cross-type roles: our Pg6 as their Bf1, our bd5 as their Rh1; transferred move g8g7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1368` (ply 29) [structural distance $d=65$, QAP obj=36]: the master played Rfd8; under μ our Rf1 plays the role of that game’s Rf1; cross-type roles: our Pg6 as their Bg7, our Ra8 as their Pg6; transferred move f8f7. Synthesis: 2 retrieved analogs transferred to Kg7 (summed QAP objective 52, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



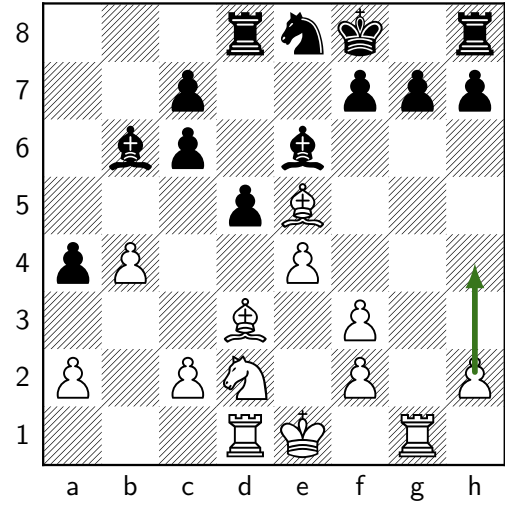
(a)



(b)

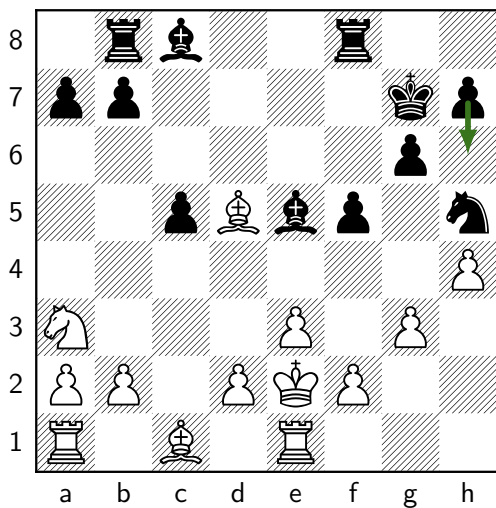


(c)

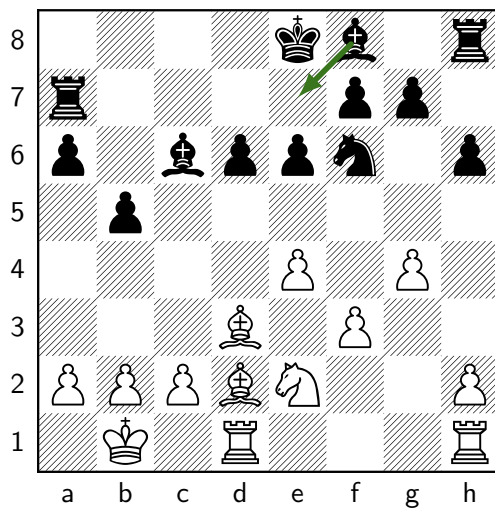


(d)

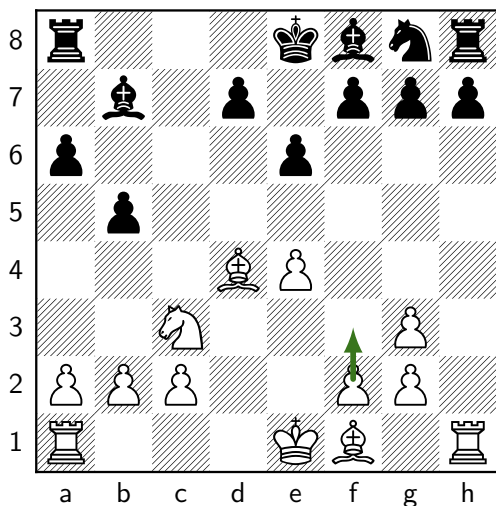
Figure 12: **Move 18, Black: Rb8.** (a) Our position; Melanie plays Rb8 (a8b8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2987` (ply 39) [structural distance $d=64$, QAP obj=74]: the master played f5; under μ our *Ra1* plays the role of that game’s *Pf2* (a **cross-type** correspondence); cross-type roles: our *ph4* as their *Ra8*, our *Ph7* as their *Rc8*; transferred move a8b8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2124` (ply 34) [structural distance $d=75$, QAP obj=66]: the master played Rc1; under μ our *Ra1* plays the role of that game’s *Ra1*; cross-type roles: our *Rf8* as their *Pa2*, our *Ph7* as their *Rd2*; transferred move a8b8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#543` (ply 38) [structural distance $d=71$, QAP obj=63]: the master played h4; under μ our *Ra1* plays the role of that game’s *Ph2* (a **cross-type** correspondence); cross-type roles: our *pa2* as their *Rd1*, our *Rf8* as their *Nd2*; transferred move a8b8 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Rb8 (summed QAP objective 344, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



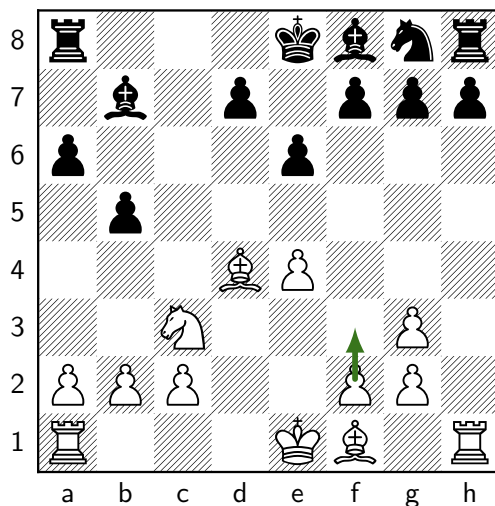
(a)



(b)

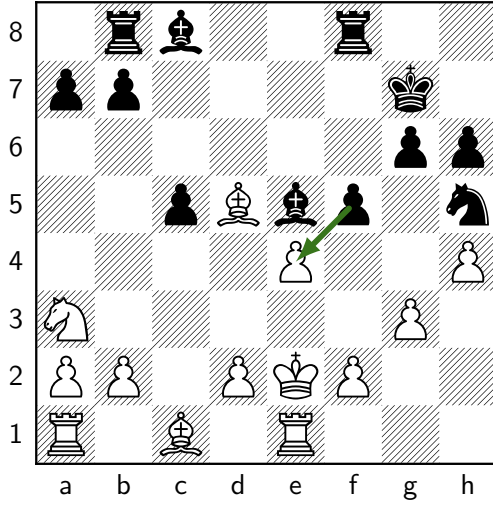


(c)

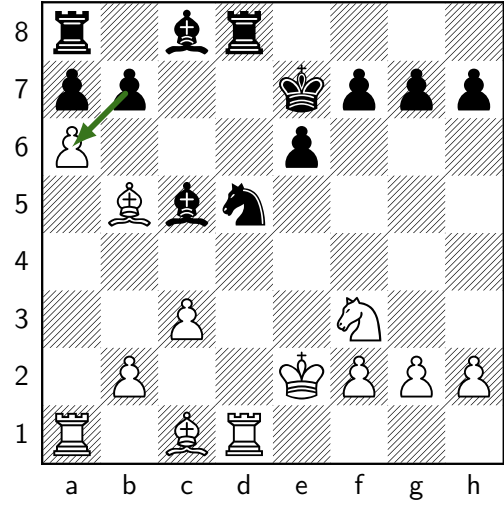


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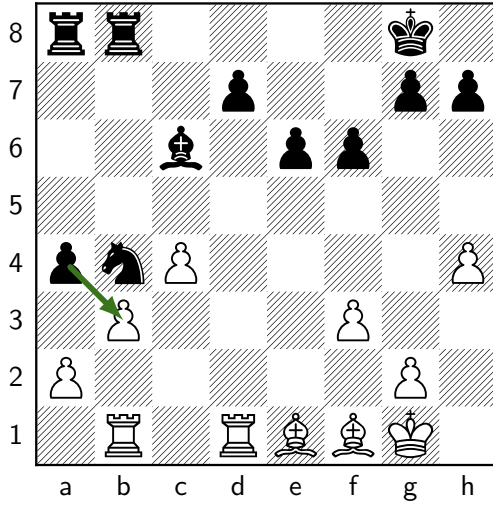
Figure 13: **Move 19, Black: h6.** (a) Our position; Melanie plays h6 (h7h6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#1584` (ply 31) [structural distance $d=69$, QAP obj=90]: the master played Be7; under μ our Ph2 plays the role of that game’s Bf1 (a **cross-type** correspondence); cross-type roles: our bc1 as their Rh8, our Rb8 as their Pa6; transferred move h7h6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#1370` (ply 20) [structural distance $d=56$, QAP obj=89]: the master played f3; under μ our Ph2 plays the role of that game’s Pf2; cross-type roles: our Pc5 as their Ra1, our Nh5 as their Bf1; transferred move h7h6 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#1485` (ply 20) [structural distance $d=56$, QAP obj=89]: the master played f3; under μ our Ph2 plays the role of that game’s Pf2; cross-type roles: our Pc5 as their Ra1, our Nh5 as their Bf1; transferred move h7h6 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to h6 (summed QAP objective 404, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



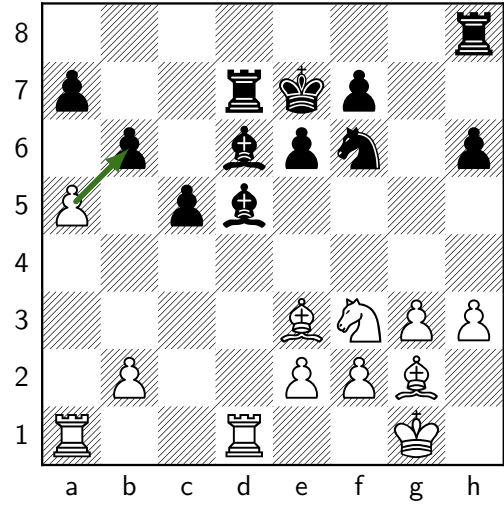
(a)



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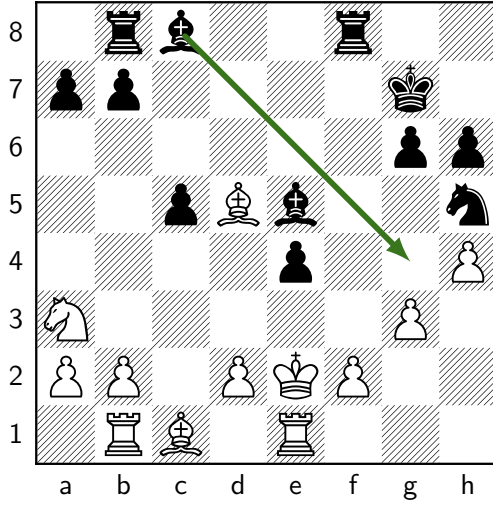


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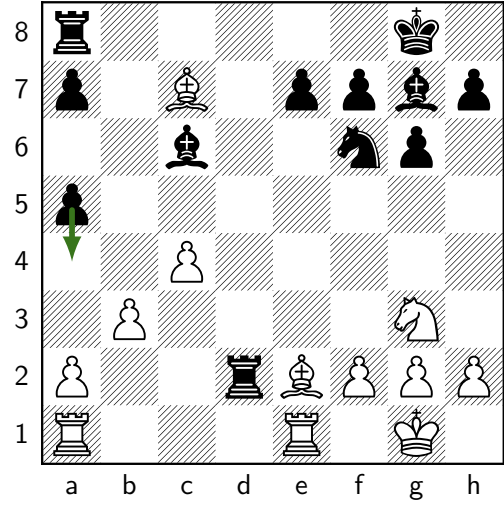


(d)

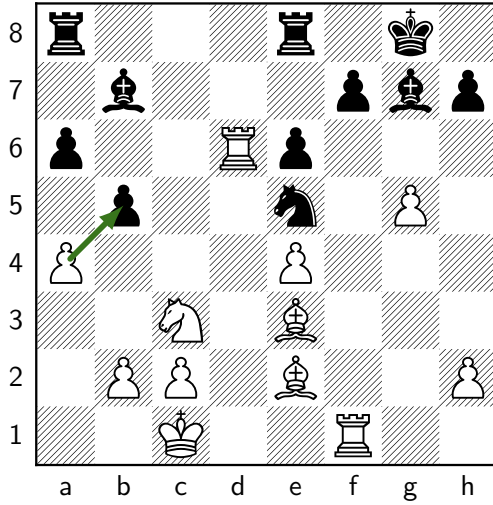
Figure 14: **Move 20, Black: fxe4**. (a) Our position; Melanie plays fxe4 (f5e4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0002#173 (ply 29) [structural distance $d=65$, QAP obj=73]: the master played bxa6; under μ our Pf4 plays the role of that game’s Pb2; cross-type roles: our Pg6 as their Bc8, our Rb8 as their Ph7; transferred move f5e4 (**endorses this move**). (c) Retrieved from ccr1_top_p0001#3502 (ply 51) [structural distance $d=64$, QAP obj=65]: the master played axb3; under μ our Pf4 plays the role of that game’s Pa5; cross-type roles: our Rb8 as their Ph7, our na3 as their Pf6; transferred move f5e4 (**endorses this move**). (d) Retrieved from ccr1_top_p0002#2142 (ply 38) [structural distance $d=63$, QAP obj=48]: the master played axb6; under μ our Pf4 plays the role of that game’s Pa5; cross-type roles: our pe4 as their Rd1, our Bc8 as their Pb2; transferred move f5e4 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to fxe4 (summed QAP objective 260, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



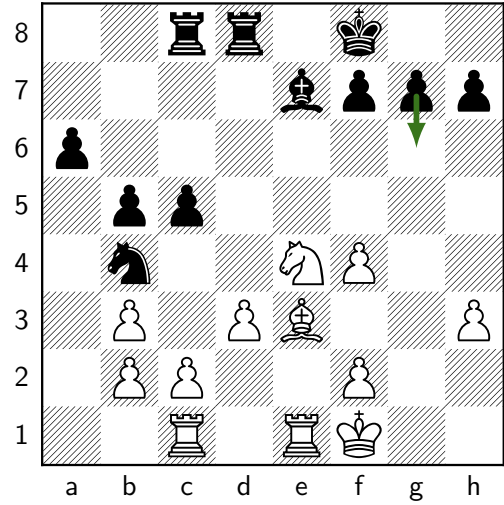
(a)



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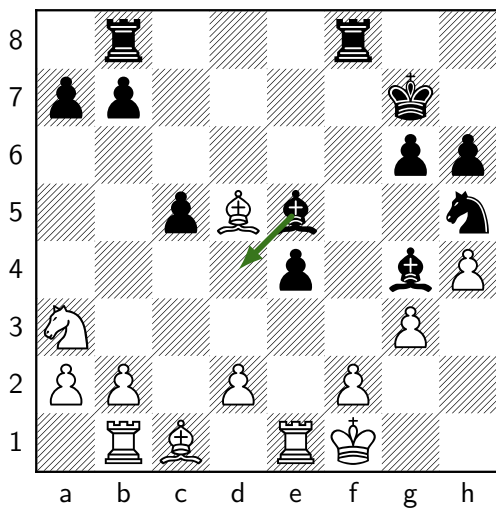


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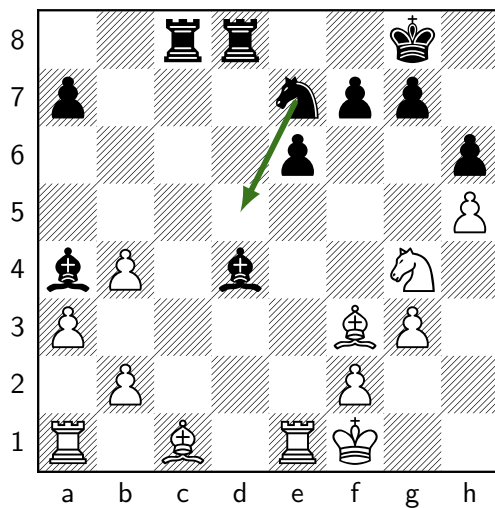


(d)

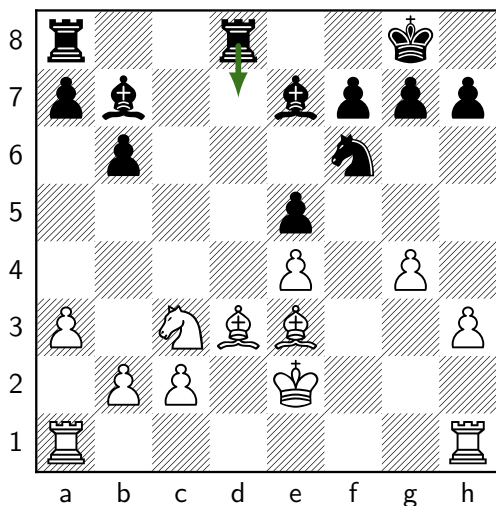
Figure 15: **Move 21, Black: Bg4+.** (a) Our position; Melanie plays Bg4+ (c8g4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3546` (ply 37) [structural distance $d=57$, QAP obj=10]: the master played a4; under μ our Bc1 plays the role of that game’s Pa4 (a **cross-type** correspondence); cross-type roles: our bc1 as their Pa7, our Ph6 as their Bg7; transferred move c8g4 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#1530` (ply 40) [structural distance $d=60$, QAP obj=54]: the master played axb5; under μ our Nh4 plays the role of that game’s Pa4 (a **cross-type** correspondence); cross-type roles: our Nh5 as their Be2, our Bc8 as their Ph2; transferred move h5g3. (d) Retrieved from `ccrl_top_p0003#2082` (ply 47) [structural distance $d=53$, QAP obj=52]: the master played g6; under μ our Bc1 plays the role of that game’s Pg2 (a **cross-type** correspondence); cross-type roles: our pd2 as their Rd8, our pf2 as their Be7; transferred move c8f5. Synthesis: 1 retrieved analog transferred to Bg4+ (summed QAP objective 10, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



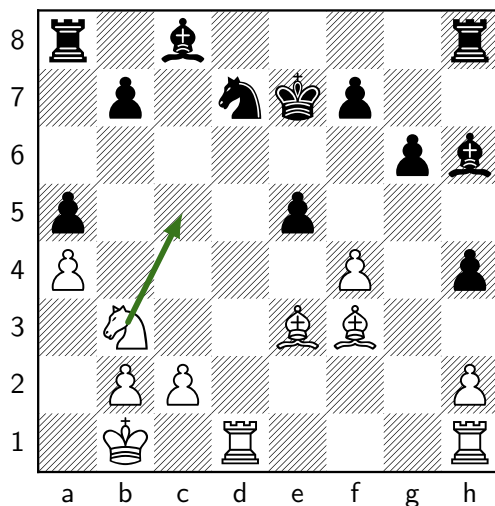
(a)



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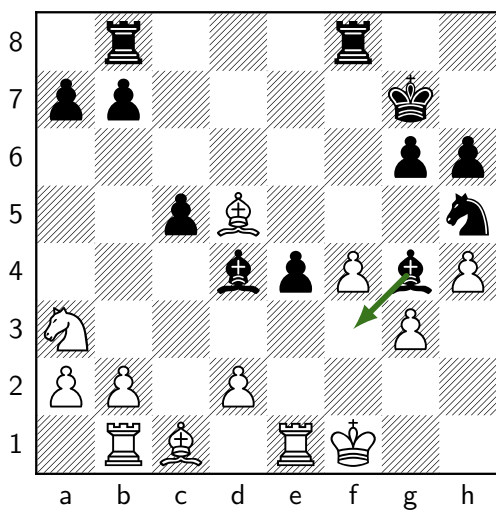


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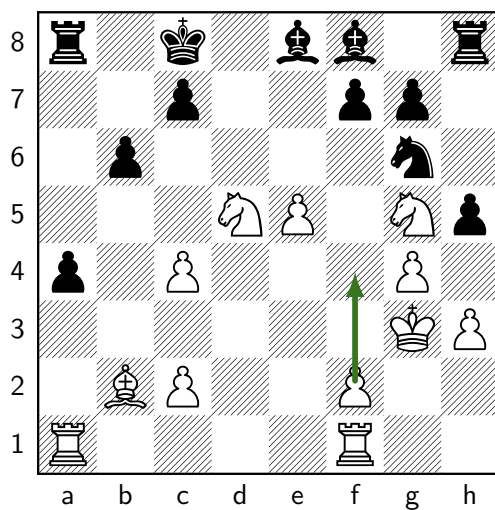


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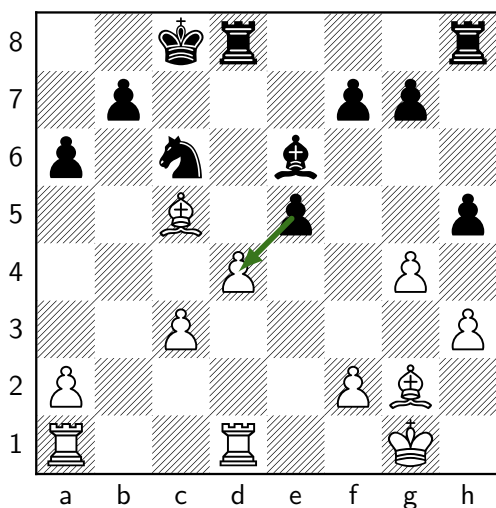
Figure 16: **Move 22, Black: Bd4.** (a) Our position; Melanie plays Bd4 (e5d4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2590` (ply 53) [structural distance $d=66$, QAP obj=53]: the master played Nd5; under μ our *Be4* plays the role of that game’s *Ne2* (a **cross-type** correspondence); cross-type roles: our *bd5* as their *Rc8*, our *bc1* as their *Pa7*; transferred move **e5d4 (endorses this move)**. (c) Retrieved from `ccr1_top_p0001#2825` (ply 29) [structural distance $d=66$, QAP obj=67]: the master played Rd7; under μ our *Nh4* plays the role of that game’s *Rd1* (a **cross-type** correspondence); cross-type roles: our *Bg4* as their *Ra8*, our *re1* as their *Bb7*; transferred move **h5g3**. (d) Retrieved from `ccr1_top_p0000#693` (ply 42) [structural distance $d=62$, QAP obj=66]: the master played Nc5; under μ our *Nh4* plays the role of that game’s *Nb3*; cross-type roles: our *Be5* as their *Rd1*, our *Pb7* as their *Rh1*; transferred move **h5g3**. Synthesis: 1 retrieved analog transferred to Bd4 (summed QAP objective 53, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



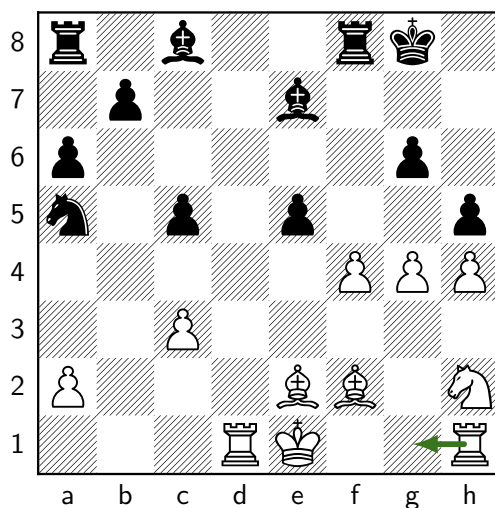
(a)



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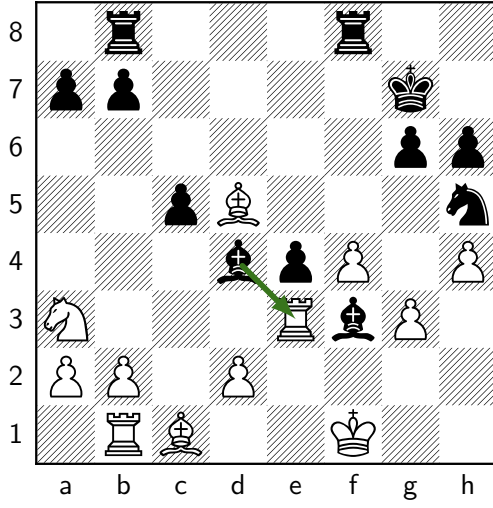


(c)

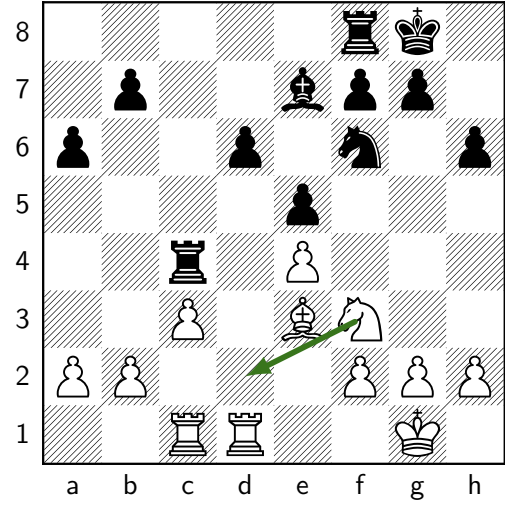


(d)

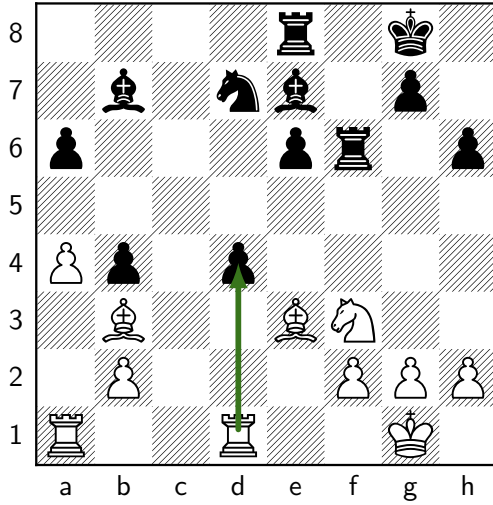
Figure 17: **Move 23, Black: Bf3**. (a) Our position; Melanie plays Bf3 (g4f3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3290` (ply 42) [structural distance $d=72$, QAP obj=90]: the master played **f4**; under μ our *Pa2* plays the role of that game’s *Pf2*; cross-type roles: our *Bd4* as their *Ng5*, our *bd5* as their *ph5*; transferred move **a7a6**. (c) Retrieved from `ccr1_top_p0002#2516` (ply 37) [structural distance $d=71$, QAP obj=87]: the master played **exd4**; under μ our *Nh4* plays the role of that game’s *Pe4* (a **cross-type** correspondence); cross-type roles: our *Bg4* as their *Pb7*, our *re1* as their *Pa6*; transferred move **h5f4**. (d) Retrieved from `ccr1_top_p0000#3300` (ply 48) [structural distance $d=71$, QAP obj=77]: the master played **Rg1**; under μ our *Pa2* plays the role of that game’s *Rh1* (a **cross-type** correspondence); cross-type roles: our *na3* as their *Rd1*, our *Rb8* as their *Be2*; transferred move **a7a6**. Synthesis: no retrieved analog transferred to Bf3 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



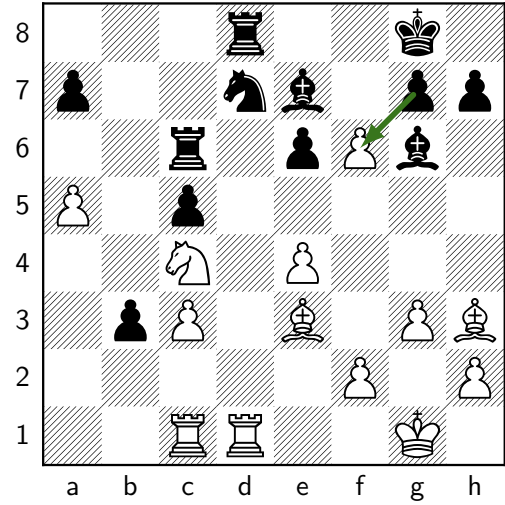
(a)



(b)

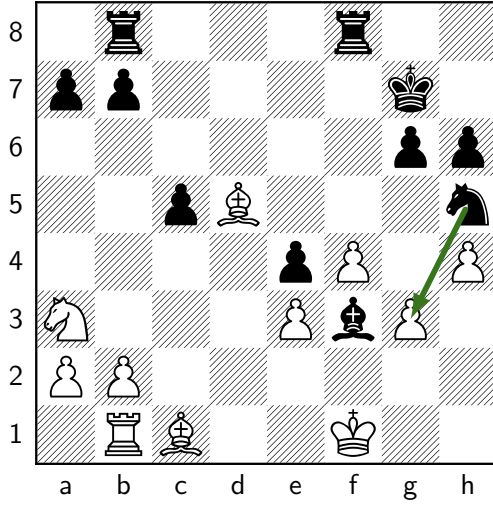


(c)

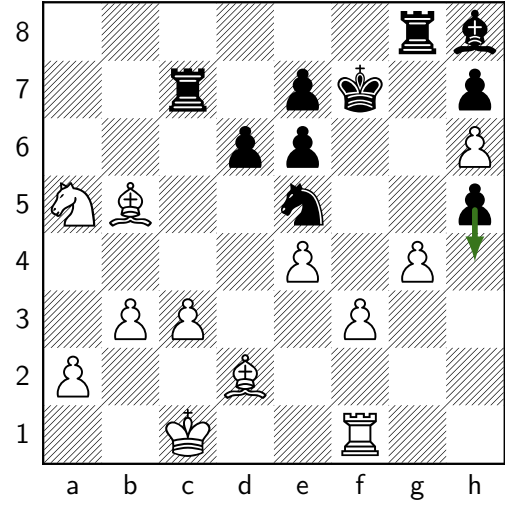


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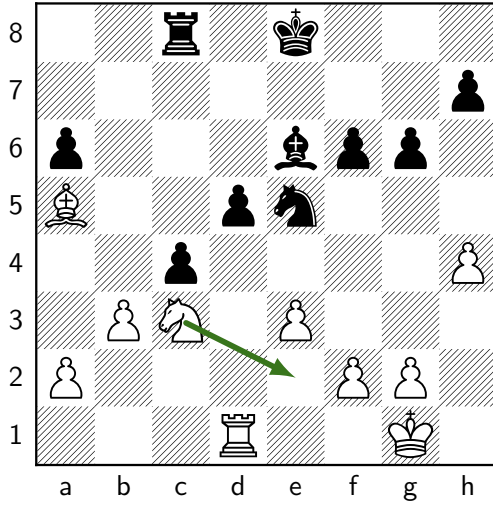
Figure 18: **Move 24, Black: Bxe3**. (a) Our position; Melanie plays Bxe3 (d4e3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0001#2245 (ply 36) [structural distance $d=90$, QAP obj=38]: the master played Nd2; under μ our Bd5 plays the role of that game’s Nf3 (a **cross-type** correspondence); cross-type roles: our Bf3 as their Rc1, our Nh5 as their Rd1; transferred move d4e3 (**endorses this move**). (c) Retrieved from ccr1_top_p0003#76 (ply 40) [structural distance $d=84$, QAP obj=38]: the master played Rxd4; under μ our Bd5 plays the role of that game’s Rd1 (a **cross-type** correspondence); cross-type roles: our bc1 as their Ra1, our Bd4 as their Nf3; transferred move d4e3 (**endorses this move**). (d) Retrieved from ccr1_top_p0001#608 (ply 43) [structural distance $d=92$, QAP obj=34]: the master played gxf6; under μ our Bd5 plays the role of that game’s Pg2 (a **cross-type** correspondence); cross-type roles: our Nh5 as their Rd8, our Bd4 as their Nd7; transferred move d4e3 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Bxe3 (summed QAP objective 115, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



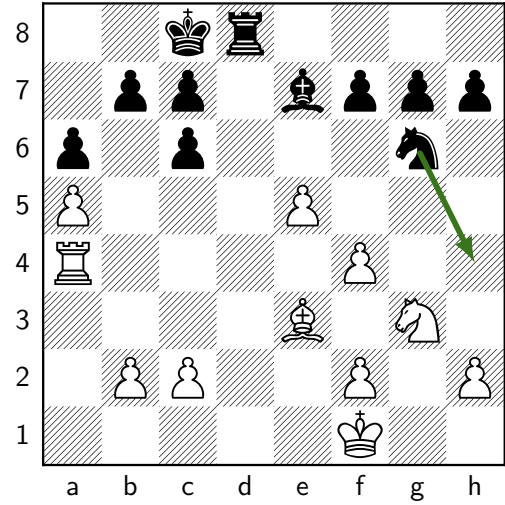
(a)



(b)



(c)



(d)

Figure 19: **Move 25, Black: Nxg3+.** (a) Our position; Melanie plays Nxg3+ (h5g3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccl1_top_p0001#1753` (ply 57) [structural distance $d=59$, QAP obj=79]: the master played **h4**; under μ our $Nh4$ plays the role of that game’s $Ph4$ (a **cross-type** correspondence); cross-type roles: our $rb1$ as their $Bh8$, our $bd5$ as their $Rc7$; transferred move **h5g3 (endorses this move)**. (c) Retrieved from `ccl1_top_p0001#176` (ply 48) [structural distance $d=69$, QAP obj=44]: the master played **Ne2**; under μ our $Nh4$ plays the role of that game’s $Nc3$; cross-type roles: our $Rb8$ as their $Pa2$, our $bd5$ as their $Pb3$; transferred move **h5g3 (endorses this move)**. (d) Retrieved from `ccl1_top_p0002#1774` (ply 39) [structural distance $d=71$, QAP obj=42]: the master played **Nh4**; under μ our $Nh4$ plays the role of that game’s $Ng3$; cross-type roles: our $Bf3$ as their $Pb7$, our $na3$ as their $Be7$; transferred move **h5g3 (endorses this move)**. Synthesis: 5 retrieved analogs transferred to Nxg3+ (summed QAP objective 230, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

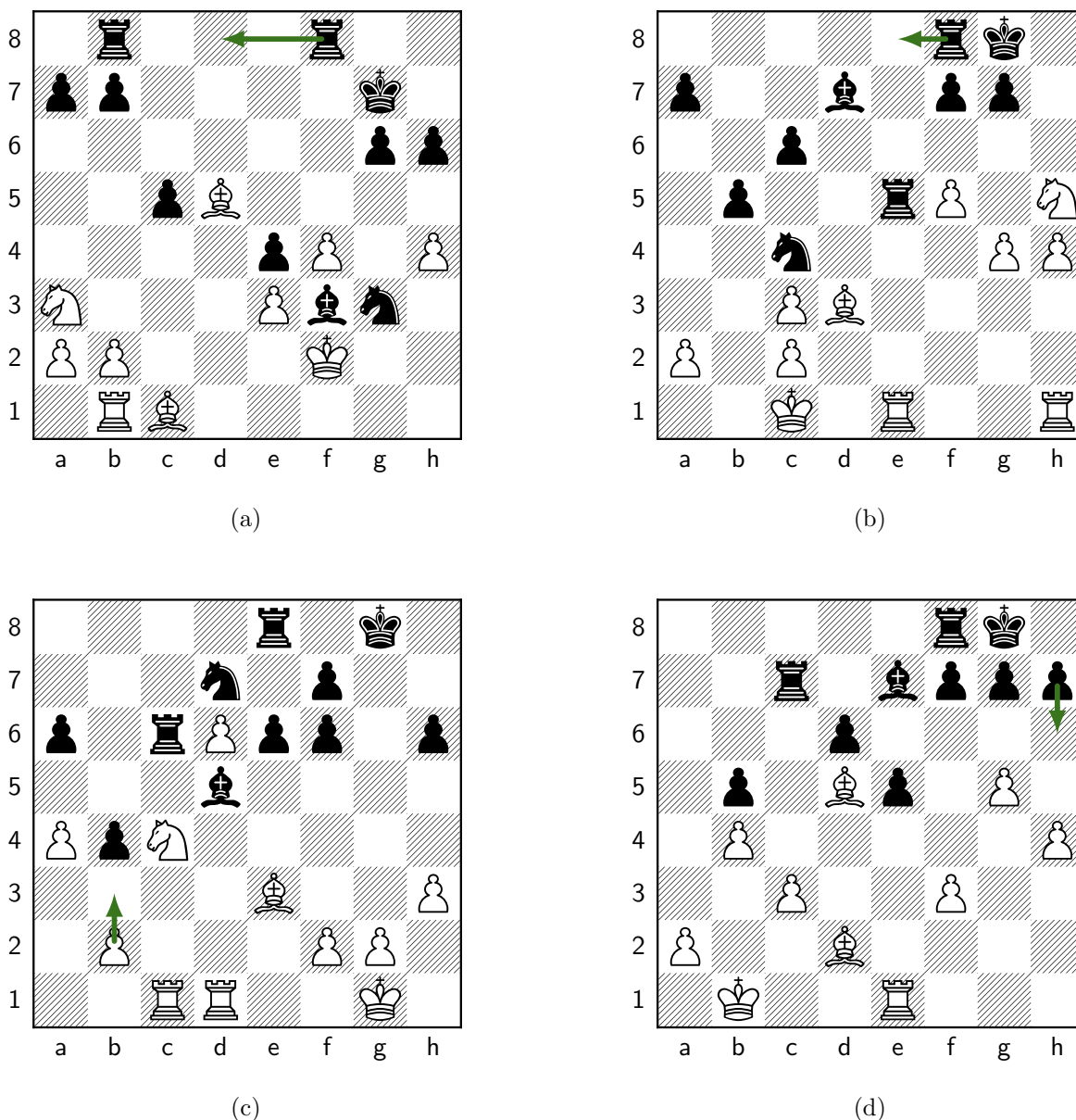


Figure 20: **Move 26, Black: Rfd8**. (a) Our position; Melanie plays Rfd8 (f8d8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#2406` (ply 49) [structural distance $d=64$, QAP obj=50]: the master played Rfe8; under μ our Rf1 plays the role of that game’s Rf1; cross-type roles: our Rb8 as their Pa7, our na3 as their Bd7; transferred move f8d8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#2058` (ply 48) [structural distance $d=71$, QAP obj=35]: the master played b3; under μ our Rf1 plays the role of that game’s Pb2 (a **cross-type** correspondence); cross-type roles: our Bf3 as their Rc1, our na3 as their Be3; transferred move f8d8 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#1375` (ply 57) [structural distance $d=67$, QAP obj=18]: the master played h6; under μ our Rf1 plays the role of that game’s Ph2 (a **cross-type** correspondence); cross-type roles: our Ng3 as their Rf8, our Bf3 as their Rc7; transferred move f8d8 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Rfd8 (summed QAP objective 103, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

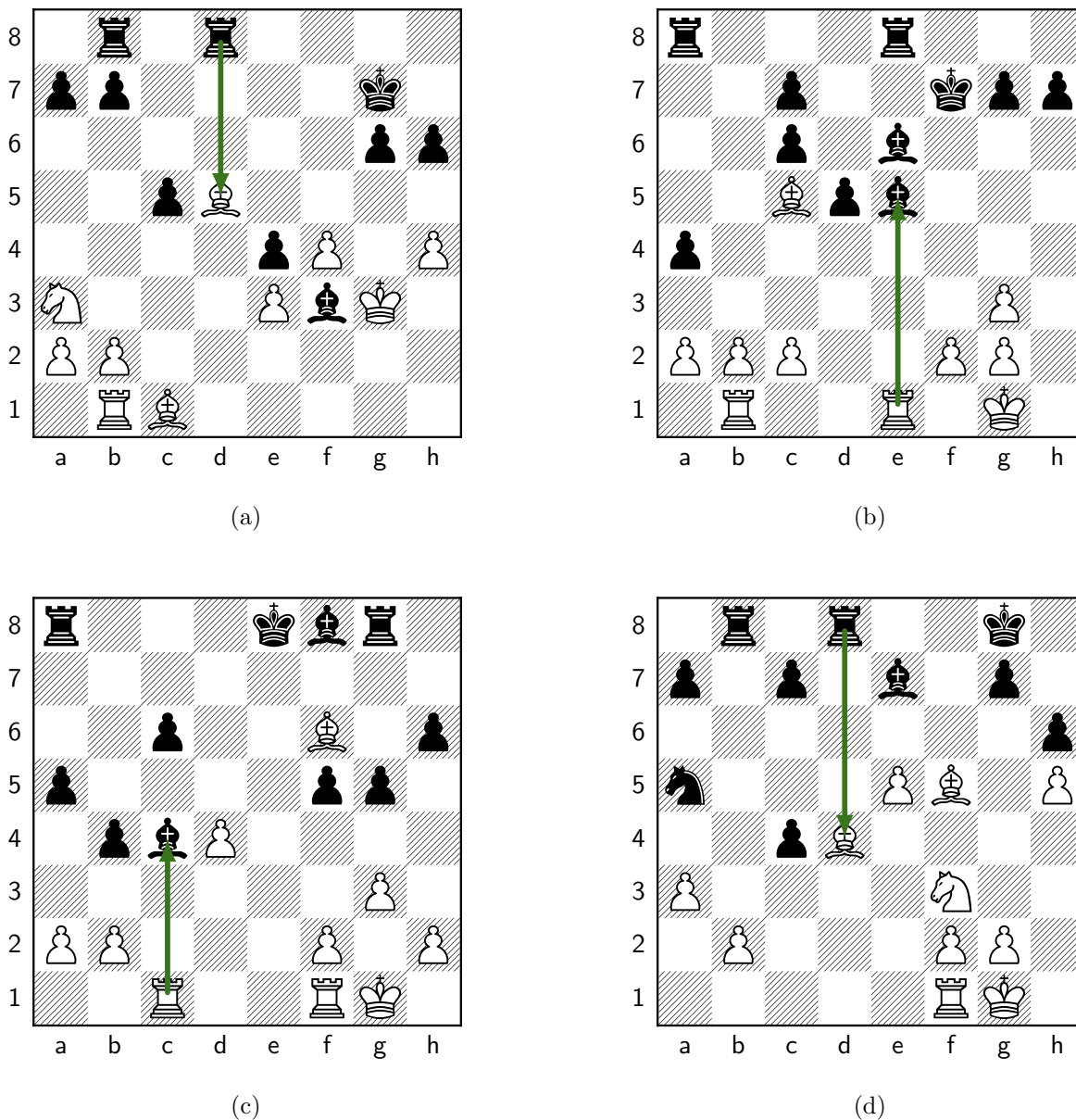
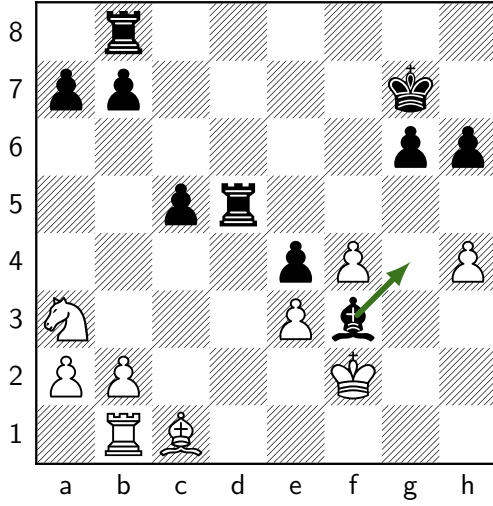
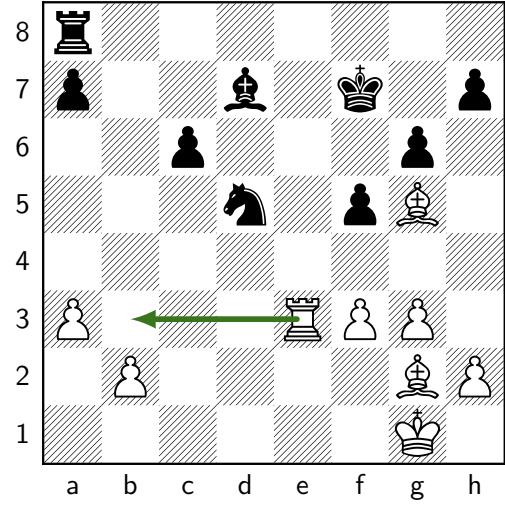


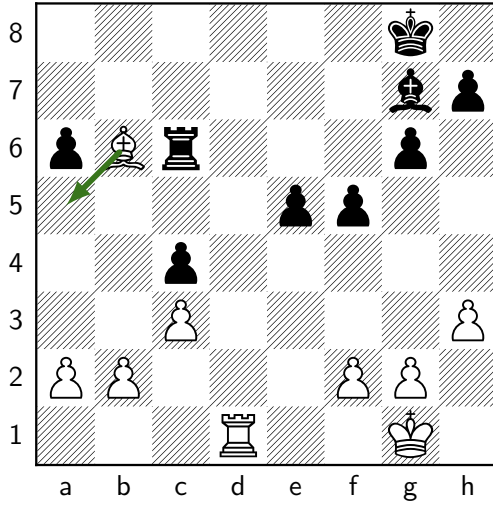
Figure 21: **Move 27, Black: Rxd5.** (a) Our position; Melanie plays Rxd5 (d8d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2527` (ply 42) [structural distance $d=53$, QAP obj=33]: the master played Rxe5; under μ our *Rd1* plays the role of that game’s *Re1*; cross-type roles: our *na3* as their *Bc5*, our *bc1* as their *pd5*; transferred move d8d5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#945` (ply 46) [structural distance $d=54$, QAP obj=22]: the master played Rxc4; under μ our *Rd1* plays the role of that game’s *Rc1*; cross-type roles: our *Pe4* as their *Rf1*, our *na3* as their *Pg3*; transferred move d8d5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3090` (ply 51) [structural distance $d=59$, QAP obj=20]: the master played Rxd4; under μ our *Rd1* plays the role of that game’s *Rd1*; cross-type roles: our *bc1* as their *Pc7*, our *pf4* as their *bf5*; transferred move d8d5 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Rxd5 (summed QAP objective 106, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



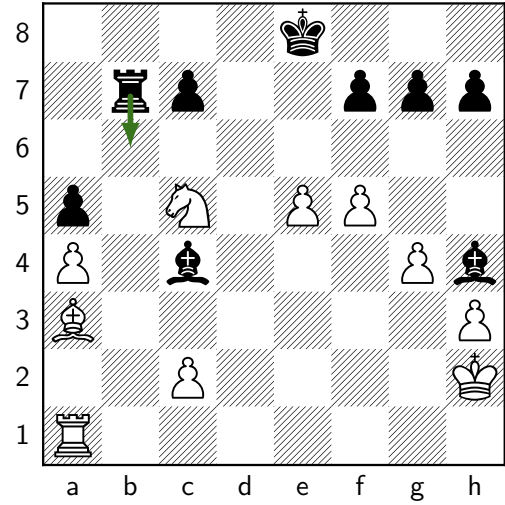
(a)



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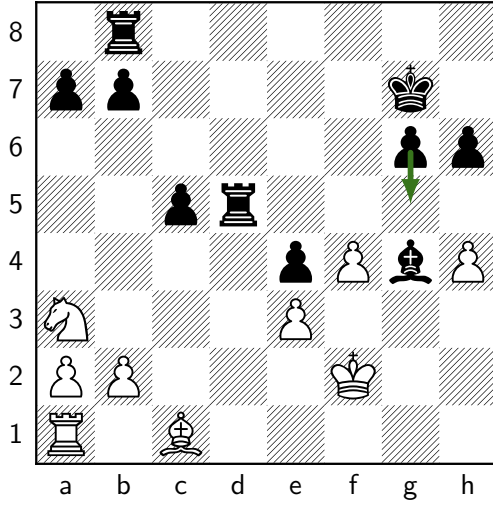


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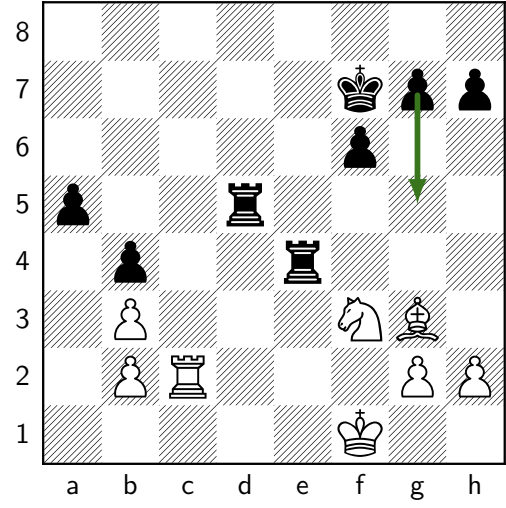


(d)

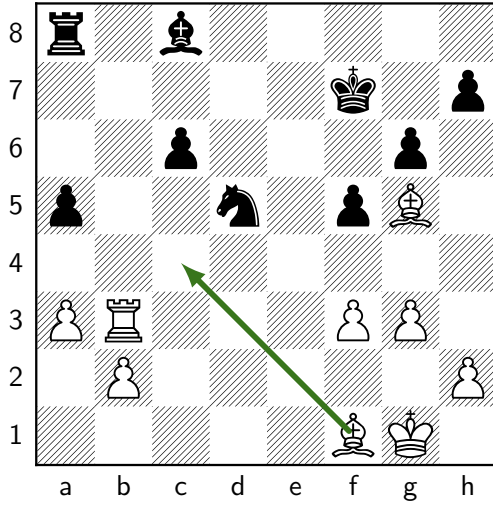
Figure 22: **Move 28, Black: Bg4**. (a) Our position; Melanie plays Bg4 (f3g4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#2356` (ply 52) [structural distance $d=45$, QAP obj=17]: the master played Rb3; under μ our Bf6 plays the role of that game’s Re3 (a **cross-type** correspondence); cross-type roles: our Rb8 as their Pb2, our Pg6 as their Bg2; transferred move f3g4 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#3425` (ply 54) [structural distance $d=44$, QAP obj=17]: the master played Ba5; under μ our Bf6 plays the role of that game’s Bb6; cross-type roles: our Rb8 as their Pg2, our na3 as their Ph3; transferred move f3g4 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#809` (ply 61) [structural distance $d=49$, QAP obj=17]: the master played Rb6; under μ our Bf6 plays the role of that game’s Rb2 (a **cross-type** correspondence); cross-type roles: our kf2 as their nc5, our na3 as their pf5; transferred move f3g4 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to Bg4 (summed QAP objective 86, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



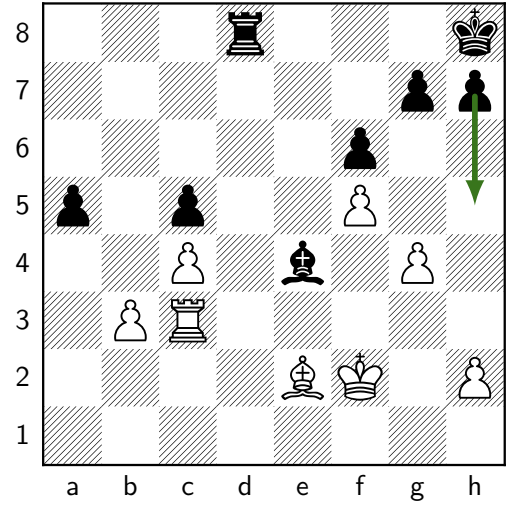
(a)



(b)

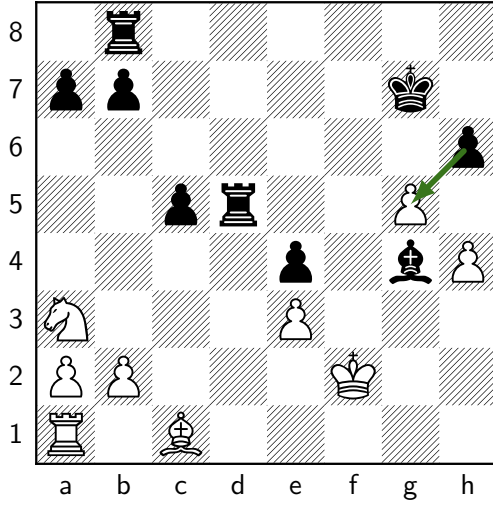


(c)

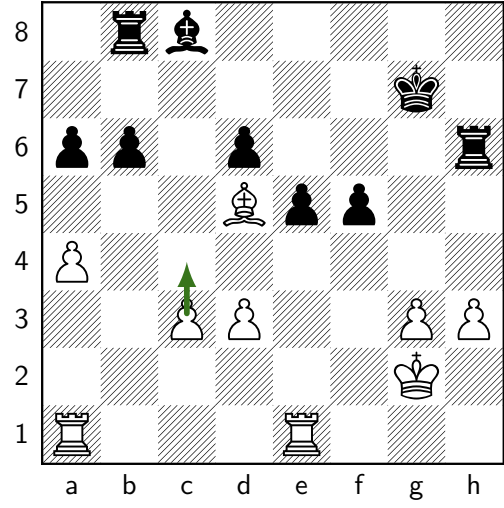


(d)

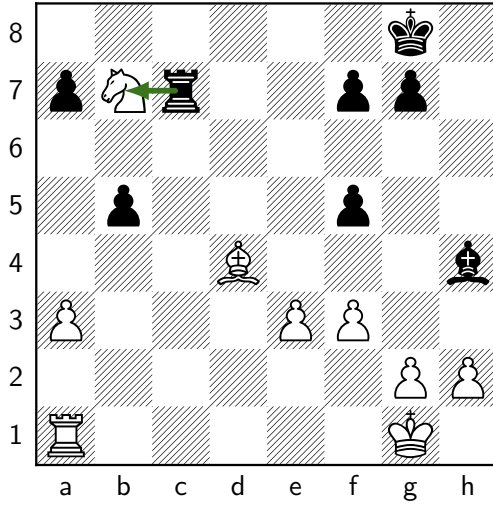
Figure 23: **Move 29, Black: g5**. (a) Our position; Melanie plays g5 (g6g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#400` (ply 65) [structural distance $d=47$, QAP obj=6]: the master played g5; under μ our $Pg3$ plays the role of that game’s $Pg2$; cross-type roles: our $Rb8$ as their $Ph7$, our $Bg4$ as their $Rd5$; transferred move g6g5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2356` (ply 56) [structural distance $d=36$, QAP obj=4]: the master played Bc4; under μ our $Pg3$ plays the role of that game’s $Bf1$ (a **cross-type** correspondence); cross-type roles: our $bc1$ as their $Pa3$, our $Rd5$ as their $Pf3$; transferred move g6g5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1581` (ply 69) [structural distance $d=46$, QAP obj=3]: the master played h5; under μ our $Pg3$ plays the role of that game’s $Ph2$; cross-type roles: our $Pc5$ as their $Rd8$, our $Bg4$ as their $Pg7$; transferred move g6g5 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to g5 (summed QAP objective 24, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



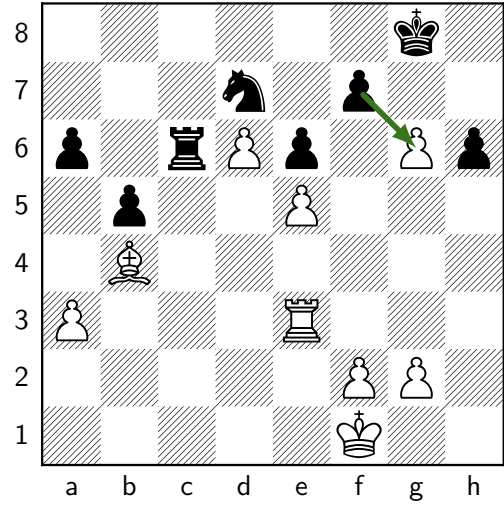
(a)



(b)

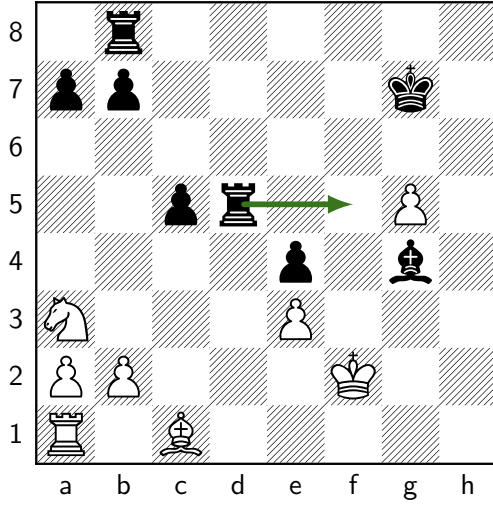


(c)

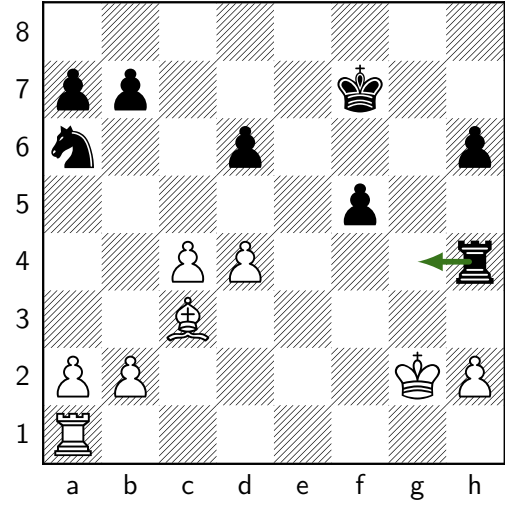


(d)

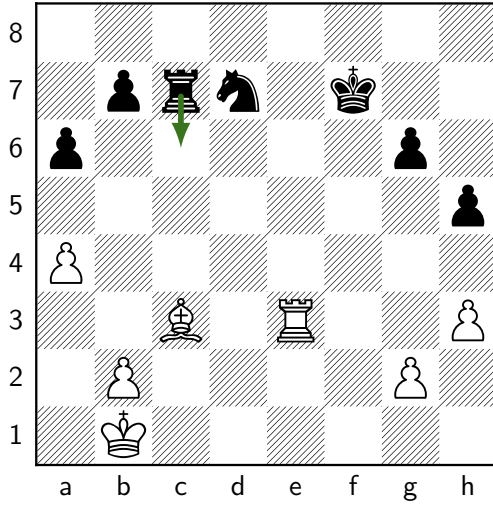
Figure 24: **Move 30, Black: hxc5**. (a) Our position; Melanie plays hxc5 (h6g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2609` (ply 56) [structural distance $d=43$, QAP obj=9]: the master played c4; under μ our *Ph3* plays the role of that game’s *Pc3*; cross-type roles: our *Pa7* as their *Ra1*, our *na3* as their *Re1*; transferred move h6g5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2321` (ply 61) [structural distance $d=43$, QAP obj=8]: the master played Rxb7; under μ our *Ph3* plays the role of that game’s *Rc2* (a **cross-type** correspondence); cross-type roles: our *pg5* as their *nb7*, our *na3* as their *Pf5*; transferred move h6g5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2840` (ply 71) [structural distance $d=41$, QAP obj=53]: the master played fxg6; under μ our *Ph3* plays the role of that game’s *Pf2*; cross-type roles: our *Rd5* as their *Nd7*, our *Ph6* as their *Rc6*; transferred move h6h5. Synthesis: 2 retrieved analogs transferred to hxc5 (summed QAP objective 17, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



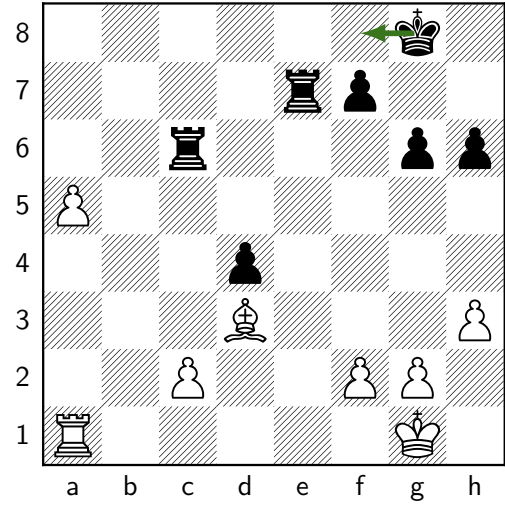
(a)



(b)

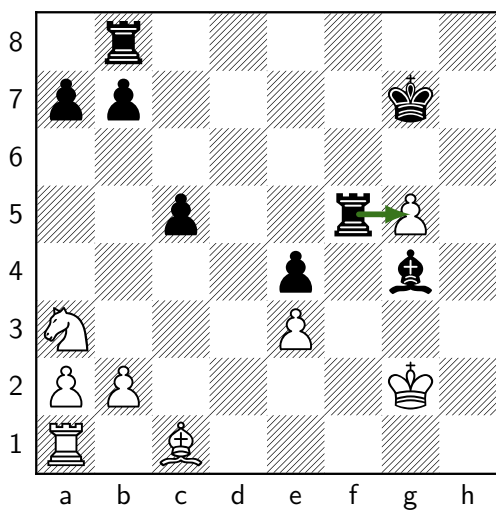


(c)

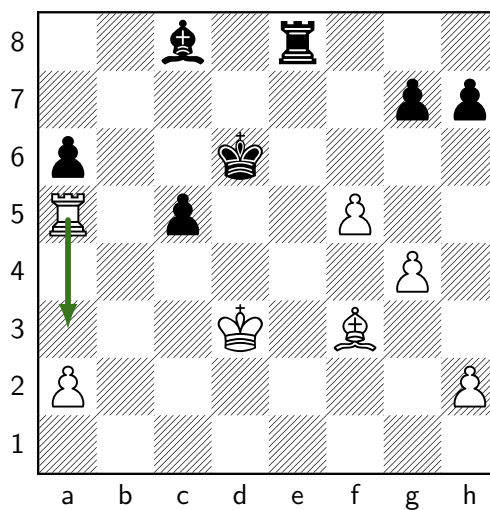


(d)

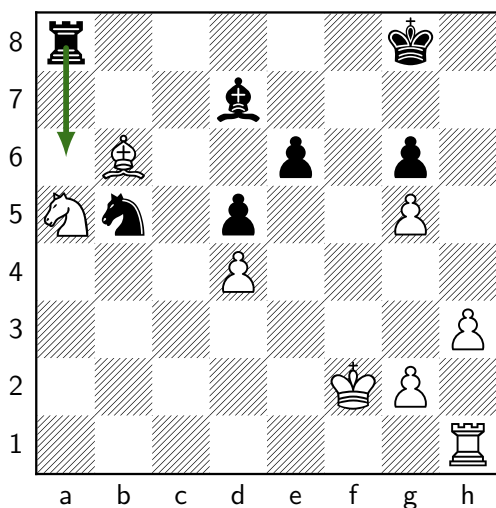
Figure 25: **Move 31, Black: Rf5+.** (a) Our position; Melanie plays Rf5+ (d5f5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3337` (ply 75) [structural distance $d=33$, QAP obj=23]: the master played Rg4+; under μ our Rd4 plays the role of that game’s Rh5; cross-type roles: our bc1 as their Pd6, our Rb8 as their Pf5; transferred move d5f5 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#1589` (ply 77) [structural distance $d=38$, QAP obj=11]: the master played Rc6; under μ our Rd4 plays the role of that game’s Rc2; cross-type roles: our Rb8 as their Nd7, our ra1 as their Pg6; transferred move d5f5 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#1855` (ply 73) [structural distance $d=36$, QAP obj=23]: the master played Kf8; under μ our Kg2 plays the role of that game’s Kg1; cross-type roles: our na3 as their Re7, our bc1 as their Pf7; transferred move g7f7. Synthesis: 4 retrieved analogs transferred to Rf5+ (summed QAP objective 62, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



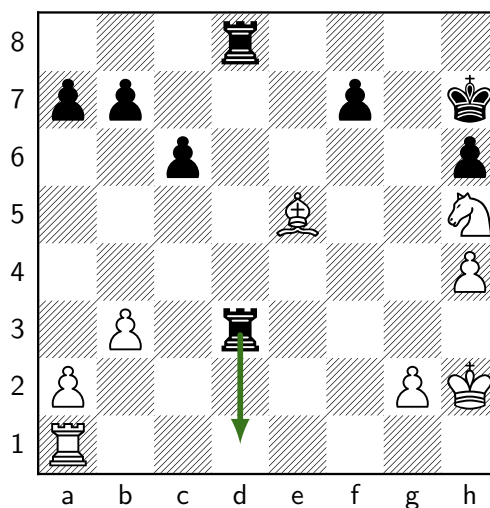
(a)



(b)

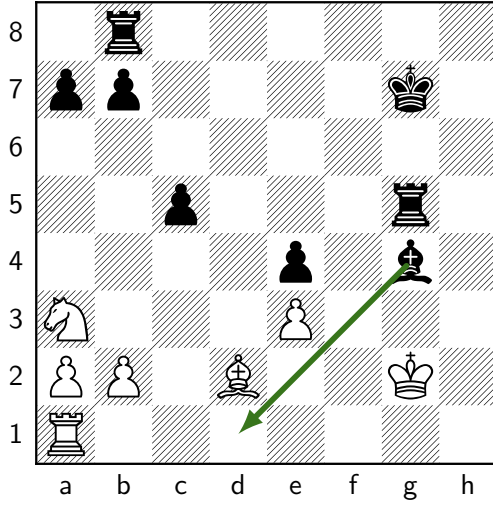


(c)

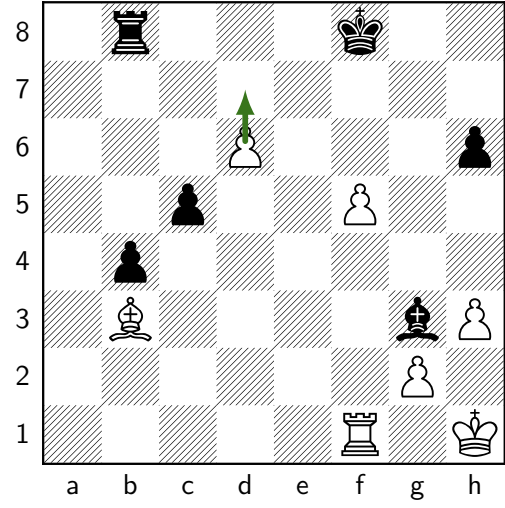


(d)

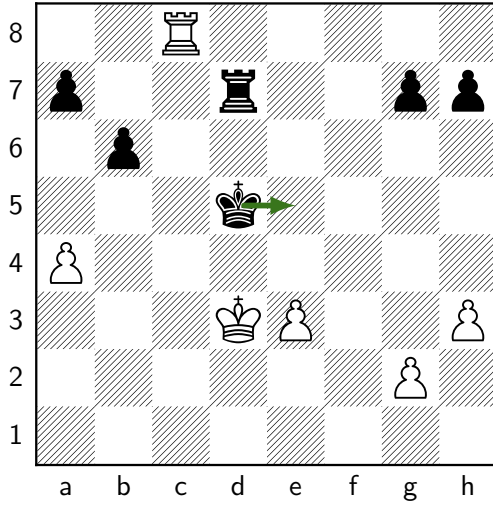
Figure 26: **Move 32, Black: Rxf5**. (a) Our position; Melanie plays Rxf5 (f5g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0001#3772 (ply 74) [structural distance $d=40$, QAP obj=27]: the master played Ra3; under μ our Rf4 plays the role of that game’s Ra5; cross-type roles: our Rb8 as their Ph2, our na3 as their Pg4; transferred move f5g5 (**endorses this move**). (c) Retrieved from ccr1_top_p0002#178 (ply 63) [structural distance $d=37$, QAP obj=19]: the master played Ra6; under μ our Rf4 plays the role of that game’s Ra1; cross-type roles: our Rb8 as their Pe6, our pg5 as their na5; transferred move f5g5 (**endorses this move**). (d) Retrieved from ccr1_top_p0000#3963 (ply 65) [structural distance $d=27$, QAP obj=11]: the master played Rd1; under μ our Rf4 plays the role of that game’s Rd6; cross-type roles: our Pc5 as their Rd8, our Rb8 as their Pa7; transferred move f5g5 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Rxf5 (summed QAP objective 94, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



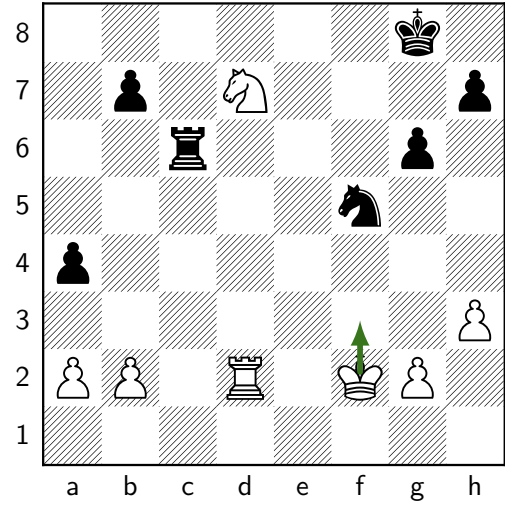
(a)



(b)

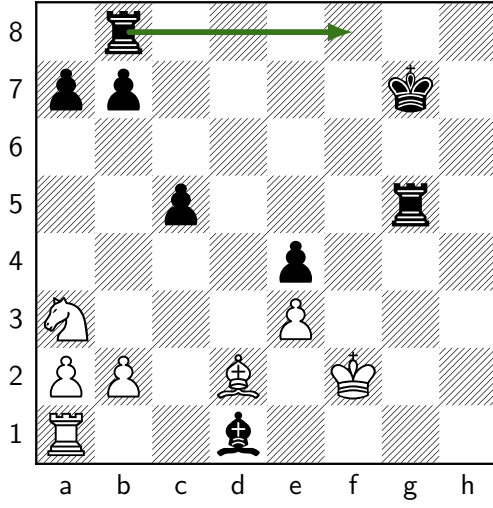


(c)

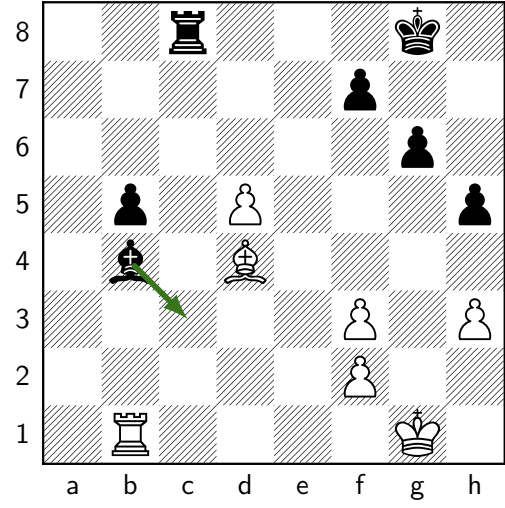


(d)

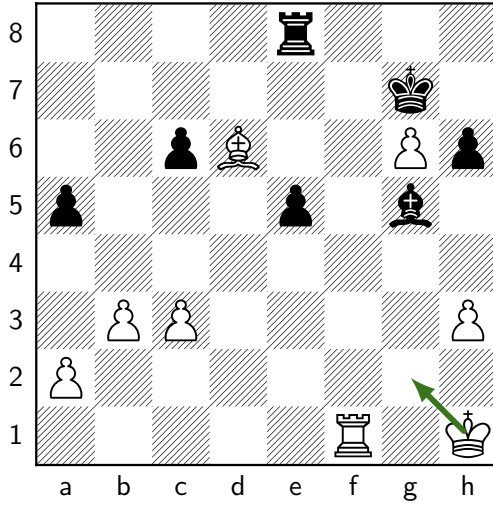
Figure 27: **Move 33, Black: Bd1+.** (a) Our position; Melanie plays Bd1+ (g4d1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#1057` (ply 68) [structural distance $d=38$, QAP obj=5]: the master played d7; under μ our Bg5 plays the role of that game’s Pd6 (a **cross-type** correspondence); cross-type roles: our na3 as their Bb3, our Rb8 as their Pf5; transferred move g4e2. (c) Retrieved from `ccrl_top_p0000#1877` (ply 67) [structural distance $d=37$, QAP obj=5]: the master played Ke5+; under μ our Kg2 plays the role of that game’s Kd4; cross-type roles: our bd2 as their rc8, our Bg4 as their Pg7; transferred move g7h7. (d) Retrieved from `ccrl_top_p0000#3822` (ply 66) [structural distance $d=34$, QAP obj=5]: the master played Kf3; under μ our Kg2 plays the role of that game’s Kf2; cross-type roles: our Rb8 as their Pg2, our bd2 as their nf5; transferred move g7f6. Synthesis: no retrieved analog transferred to Bd1+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



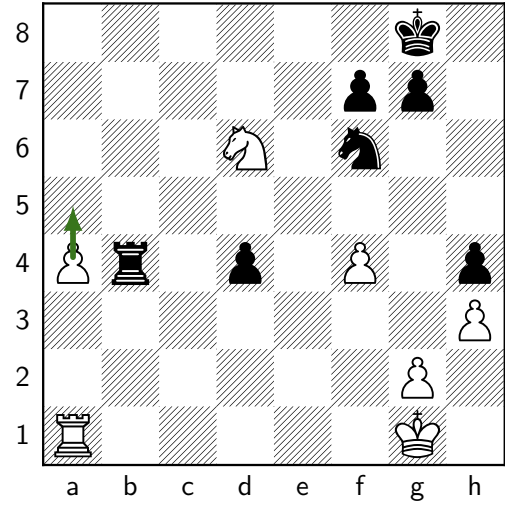
(a)



(b)

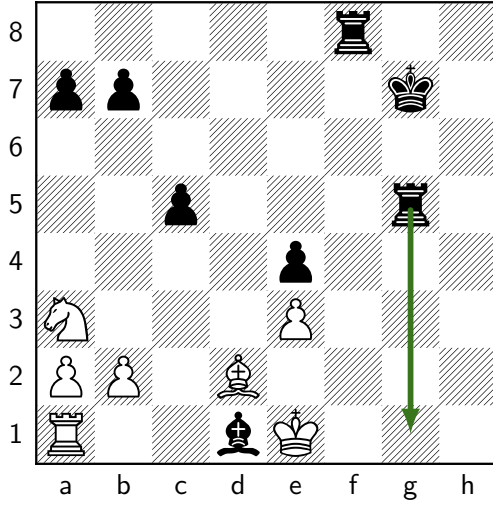


(c)

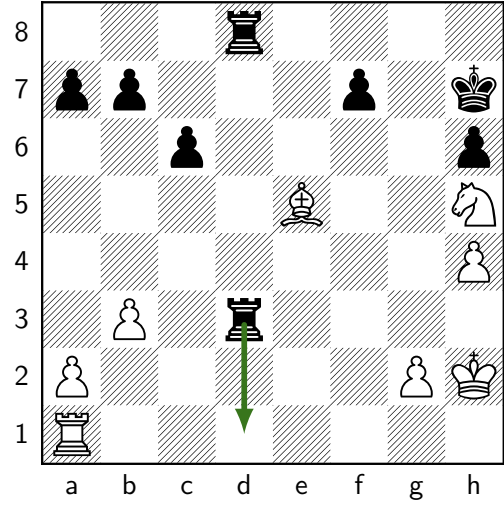


(d)

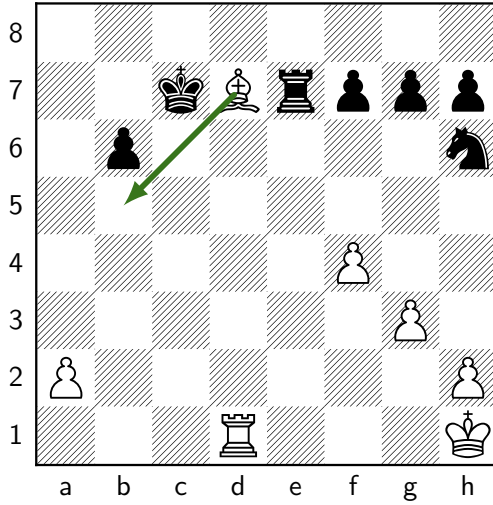
Figure 28: **Move 34, Black: Rf8+.** (a) Our position; Melanie plays Rf8+ (b8f8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#224` (ply 73) [structural distance $d=35$, QAP obj=21]: the master played Bc3; under μ our Bd8 plays the role of that game’s Bb5; cross-type roles: our na3 as their Pb5, our bd2 as their Ph5; transferred move d1c2. (c) Retrieved from `ccr1_top_p0000#2501` (ply 72) [structural distance $d=35$, QAP obj=21]: the master played Kg2; under μ our Kg2 plays the role of that game’s Kh1; cross-type roles: our na3 as their pa5, our Bd1 as their pe5; transferred move g7f7. (d) Retrieved from `ccr1_top_p0000#2423` (ply 66) [structural distance $d=34$, QAP obj=21]: the master played a5; under μ our Bd8 plays the role of that game’s Pa4 (a **cross-type** correspondence); cross-type roles: our na3 as their Pf4, our Rb8 as their Nd6; transferred move d1c2. Synthesis: no retrieved analog transferred to Rf8+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



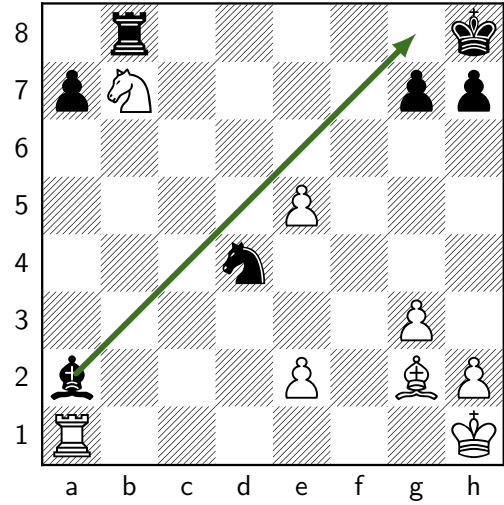
(a)



(b)



(c)



(d)

Figure 29: **Move 35, Black: Rg1#.** (a) Our position; Melanie plays Rg1# (g5g1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3963` (ply 65) [structural distance $d=33$, QAP obj=2]: the master played Rd1; under μ our Rg4 plays the role of that game’s Rd6; cross-type roles: our Pc5 as their Rd8, our Bd1 as their nh5; transferred move g5g1 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#185` (ply 62) [structural distance $d=37$, QAP obj=38]: the master played Bb5; under μ our Bd8 plays the role of that game’s Bd7; cross-type roles: our bd2 as their pb6, our ke1 as their re7; transferred move d1g4. (d) Retrieved from `ccrl_top_p0000#1815` (ply 57) [structural distance $d=32$, QAP obj=26]: the master played Bg8; under μ our Bd8 plays the role of that game’s Ba7; cross-type roles: our ke1 as their Rb8, our Rg5 as their Pa7; transferred move d1h5. Synthesis: 1 retrieved analog transferred to Rg1# (summed QAP objective 2, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.